

CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL VALLEY REGION

ORDER NO. R5-2002-0062

WASTE DISCHARGE REQUIREMENTS
FOR
U. S. DEPARTMENT OF NAVY
LEMOORE NAVAL AIR STATION
WASTEWATER TREATMENT FACILITY
KINGS COUNTY

The California Regional Water Quality Control Board, Central Valley Region, (hereafter Regional Board) finds that:

1. The U.S. Department of Navy Lemoore Naval Air Station (hereafter LNAS or Discharger) owns and operates an industrial wastewater treatment facility (WWTF) for treating wastewater from washing aircrafts and a domestic WWTF that provides sewage service for about 15,000 residents. Effluent from the industrial WWTF is discharged to the domestic WWTF. The WWTFs are about six miles west of the City of Lemoore. The industrial WWTF is in Section 31, T18S, R19E, MDB&M; the domestic WWTF is in Sections 26, 27, 34, and 35, T19S, R19E, MDB&M. Attachment A, a part of this Order, shows the locations of both WWTFs.
2. The Regional Board adopted Waste Discharge Requirements (WDRs) Order No. 86-173 for the Discharger's industrial and domestic WWTFs. It prescribed requirements for a total discharge from the domestic WWTF to evaporation ponds of 2.12 million gallons per day (mgd), of which up to 0.1 mgd may be from the Discharger's industrial WWTF. Current discharge flow is about 1.9 mgd.
3. Order No. 86-173 set industrial effluent limitations for constituents of concern based on drinking water standards, State and EPA health advisory levels, and EPA best available technology limitations. Specifically, Order No. 86-173 prescribed industrial WWTF effluent limitations for total suspended solids (TSS), settleable solids, copper, nickel, total chromium, zinc, total cyanide, silver, lead, cadmium, trichloroethane, oil and grease, and conductivity at 25° Celsius (EC). Order No. 86-173 is subject to and due for periodic review. The purpose of this Order is to rescind the previous Order and update waste discharge requirements, in part, to ensure the discharge is consistent with Regional Board plans and policies and to prescribe requirements for protecting existing and potential beneficial uses of receiving waters.
4. The Discharger's sewer collection system consists of two main sections: (a) the Operations Area, which conveys industrial effluent and about 0.2 mgd of domestic wastewater; and (b) the Administrative and Housing Area, which conveys about 1.8 mgd of domestic wastewater. A six-mile long pressurized pipeline conveys Operations Area wastewater to the Administrative and Housing Area. Attachment D, a part of this Order, shows the location of the Operations Area and the Administrative and Housing Area. The Discharger indicates that it has replaced about two-thirds of its sewer collection system with new PVC pipe. The remaining portions of the sewer collection system consist of old red clay pipe.
5. About 90 percent of the wastewater discharged to the industrial WWTF consists of washdown from aircraft washing. The remaining wastewater discharged to the industrial WWTF includes

washdown from jet engine test cells and bearings and brake shops. The industrial WWTF includes oil and water separators, storage tanks, dissolved air flotation units, and sedimentation tanks. After treating the industrial wastewater, LNAS pumps it to the main lift station where it combines with raw domestic wastewater and filter backwash water from the Discharger's onsite water treatment plant. The domestic WWTF includes two "muffin monsters," two stabilization ponds (Ponds A and B), and three evaporation ponds (Ponds 1, 2, and 3), which total 340 acres. Attachments B and C, a part of this Order, depict the industrial WWTF and domestic WWTF flow diagrams, respectively.

6. The Discharger adjusts the pH of industrial effluent with sulfuric acid prior to discharging it to a six-mile force main that also conveys about 0.2 mgd of domestic wastewater. As domestic wastewater will effectively neutralize the pH of industrial effluent and the Discharger indicates that the piping containing only industrial effluent is compatible with high pH, it is not necessary for this Order to prescribe a pH limitation for industrial effluent.
7. Hydraulic conductivity of soils underlying the domestic WWTF ponds ranges from 5×10^{-9} to 7.3×10^{-7} cm/sec, according to laboratory permeability tests performed by the Discharger in 1971. Extensive clay layers in the area of the domestic WWTF ponds result in perched groundwater (depth to groundwater is about 10 feet below ground surface). In response to concerns of neighboring farmers that effluent was adversely affecting farming activities (i.e., creating drainage problems), LNAS installed a leachate collection system to collect any effluent that may be percolating from its stabilization and evaporation ponds. To dispose of leachate, LNAS discharges it to a dedicated evaporation pond (Pond 3). A portion of the interceptor drain surrounding the ponds is broken, but the Discharger indicates it plans to replace the entire interceptor drain.
8. Source Water. The LNAS obtains its source water from the California Aqueduct. Source water quality is good, as indicated by the LNAS's 2001 Annual Water Quality Report. Excerpts of this report are presented below.

<u>Constituent / Parameter</u>	<u>Units</u>	<u>Value</u>
EC	µmhos/cm	437
Total Dissolved Solids (TDS)	mg/L	238
Sodium	mg/L	47
Chloride	mg/L	71
Nitrate (as N)	mg/L	0.35

9. Self-monitoring data from January 2000 to October 2001 characterize industrial WWTF effluent as follows:

<u>Constituent / Parameter</u>	<u>Units</u>	<u>Average</u>	<u>Maximum</u>
Flow	mgd	0.014	0.092
Oil and Grease	mg/L	51.5	230
TSS	mg/L	15.9	80
EC	µmhos/cm	1,070	2,010

10. Order No. 86-173 required the Discharger to monitor influent flow and dissolved oxygen levels in its two stabilization ponds and three evaporation ponds. On 24 October 2001, staff collected samples of influent (prior to the two “muffin monsters”) and effluent (after the second stabilization pond where the Discharger routes wastewater from Pond A to Pond 1). These results characterize the composition of domestic WWTF influent and effluent as follows:

<u>Constituent / Parameter</u>	<u>Units</u>	<u>Influent</u>	<u>Effluent</u>
pH	pH units	7.5	8.5
BOD ₅ ¹	mg/L	62	13
TSS	mg/L	200	27
EC	µmhos/cm	6,700	14,000
TDS	mg/L	5,200	11,000
Chloride	mg/L	450	1,000
Sulfate	mg/L	2,800	5,800
Calcium	mg/L	95	170
Magnesium	mg/L	140	320
Potassium	mg/L	22	47
Sodium	mg/L	1,300	3,000
Bicarbonate	mg/L	55	320
Nitrate-nitrogen	mg/L	1.2	< 0.1
Ammonia-nitrogen	mg/L	10	4.9
Boron	mg/L	5.5	11
Selenium	µg/L	NA	< 2

¹Five-day biochemical oxygen demand

11. The high concentrations of sodium and sulfate in the domestic WWTF influent and effluent are indicative of the presence of tile drain water (described in Finding No. 22). In addition, the low concentration of BOD₅ (i.e., 62 mg/L) in the domestic WWTF influent indicates that domestic wastewater has been significantly diluted. Typically, untreated domestic wastewater contains BOD₅ concentrations ranging from 110 to 400 mg/L, according to *Wastewater Engineering* by Metcalf & Eddy. To explain the high levels of EC in its discharge, the Discharger indicated that construction activities it began at the base around July 2001 require significant dewatering. The Discharger also indicated that it expects significant dewatering to continue until about June 2002 when it will complete construction of an eight-foot diameter pipe for conveying storm water.
12. During a 24 October 2001 inspection, the Discharger indicated that the parshall flume used to measure flow at its domestic WWTF did not provide accurate flow measurement until the Discharger recalibrated it in October 2000. Self-monitoring reports from January 2000 through October 2001 indicate that the Discharger reported average daily flows from January through October 2000 of about 0.8 mgd and from November 2000 to October 2001 of about 1.9 mgd.
13. Self-monitoring data also indicates that daily flows to the domestic WWTF can vary significantly. For example, the June 2001 self-monitoring report indicated a range of daily flows between 0.68 and 2.49 mgd. As domestic WWTFs typically receive a similar quantity of wastewater each day, it appears that either the discharge of shallow groundwater (resulting from dewatering

associated with base construction), infiltration of tile drain water into the wastewater collection system, or both, significantly effect the quantity of wastewater treated by the domestic WWTF. The average per capita domestic use (including public service) of water typically generates about 70 gallons of wastewater per day, according to *Wastewater Engineering* by Metcalf & Eddy. For the 15,000 residents that live onsite this should equate to an average daily domestic wastewater flow of about 1.05 mgd. Accordingly, it appears that low flow values reported in self-monitoring reports are indicative of days when infiltration of tile drain water into the collection system and dewatering from onsite construction were insignificant.

14. Dissolved Oxygen Monitoring. To prevent nuisance conditions and to reduce the likelihood of an avian botulism outbreak, the Discharger needs to ensure that dissolved oxygen levels in its domestic WWTF ponds remain above 1 mg/L. This is because the microorganism that causes avian botulism prefers anaerobic conditions. The appropriate mechanism for ensuring that nuisance conditions do not develop is to require dissolved oxygen monitoring in the early morning (the minimum level for a diurnal variation). The Discharger has not reported the time it collects samples in self-monitoring reports and self-monitoring data from January 2000 through October 2001 that indicates dissolved oxygen levels ranging from 0.0 to 24.0 mg/L is suspect.
15. Compliance Issues. Self-monitoring data from January 2000 through October 2001 indicate that the Discharger chronically violated industrial effluent limitations for EC and oil and grease. In October 2001, the Discharger also violated industrial effluent limitations for nickel and cadmium. Self-monitoring data from 2000 and 2001 indicate that dichloromethane was detected in industrial effluent in January 2001 (460 µg/L), June 2001 (5.5 µg/L), and December 2001 (6.8 µg/L). Title 22, California Code of Regulations (CCR), section 64444 establishes a drinking water maximum contaminant level for dichloromethane of 5 µg/L.

Sludge Management

16. Industrial Sludge Management. Sludge from the holding tank, oil and water separator and sedimentation basin is discharged to a sludge holding tank, then to a 1,200-gallon metal diatomaceous earth tank before being dewatered by a pressure filter press. All leachate and decant are recycled back to the headworks of the industrial WWTF. The spent diatomaceous earth and sludge are temporarily stored in metal bins before being landfilled at Chemical Waste Management's Kettleman Hills Facility, which the Regional Board regulates by Order No. 98-058.
17. Municipal Sludge Management. The Discharger has not removed sludge from its domestic WWTF since it became operational in 1975. In 1995, the Discharger determined that about 0.5 feet of sewage sludge had accumulated in each of the five ponds comprising its domestic WWTF, which equates to a sludge build-up rate of about 0.3 inch/year. For sludge to accumulate to levels that would affect the treatment process, it would take sixty years, as stabilization ponds at the domestic WWTF are reportedly capable of providing adequate treatment at a minimum depth of 4.5 feet, according to the Discharger's 1995 *Sewage Sludge Management Plan*.

Hydrology, Soils and Land Use

18. The WWTF lies within the Tulare Lake Basin, specifically the Hanford-Lemoore Hydrologic Area (No. 551.90) as depicted on interagency hydrologic maps prepared by the California Department

of Water Resources (DWR) in 1986. Areal topography indicates a slope of about 1 foot per 1,000 feet toward the east. Surface water drainage in the WWTF environs is by sheet flow to the Kings River, approximately two miles to the east, close to where the North and Clarks Fork converge.

19. The area is characterized by warm, dry summers and cool, wet winters. The average annual rainfall is 8 inches, 80 percent of which occurs between November and March. The average annual evapotranspiration rate is 63 inches per year.
20. *Final Basewide Groundwater Beneficial Use Technical Memorandum* (hereafter Report), dated 6 April 2001, and prepared by Tetra Tech EM Inc. for the US Department of Navy provides the following information for LNAS: Beneath the LNAS facility, which encompasses about 20 square miles, the A-Clay can be found at about 50 feet below ground surface (bgs), the C-Clay at about 225 feet bgs, and the E-Clay at about 650 feet bgs. The E-Clay separates the upper and lower aquifers. The shallow-upper aquifer is found above 50 feet bgs, the intermediate-upper aquifer, between 50 and 225 feet bgs, and the deep-upper aquifer, between 225 and 650 feet bgs.
21. The Report indicates that soils to the E-Clay consist of clay and alluvial deposits with a small amount of fluvial deposits. Alluvial deposits consist of gravel, sand, and fine-grained sediments; while fluvial deposits are predominantly gravel, coarse to fine-grained sand, silt, and clay. As several extensive clay layers exist in the upper aquifer system, the permeability of area soils is considered low. The Report indicates that shallow groundwater is typically 9 to 15 feet bgs.
22. The Report evaluates water quality in the shallow-upper aquifer and concludes that it cannot potentially be used for either municipal or agricultural supply due to high TDS levels. The primary source of high TDS is irrigated agriculture. As the permeability of soils near LNAS prevents the downward movement of salts, surrounding farmers utilize interceptor drains to dewater shallow groundwater and reduce salt accumulation in the soil. Typically, farmers discharge the tile drain water to evaporation basins under Regional Board-adopted waste discharge requirements.
23. The Report provides stiff diagrams from cluster wells that are screened at multiple intervals to show the geochemical difference in water quality below the shallow-upper aquifer to illustrate that the A-Clay limits the vertical transport of water. The Report indicates that the absence of agricultural wells in the shallow-upper and intermediate-upper aquifer and the high TDS levels in both of these aquifers show that they are not suitable for agricultural supply.
24. The Report indicates that LNAS operations have degraded groundwater at several sites throughout the base. The attached Information Sheet describes these operations and the effect they have had on groundwater.
25. Between 1985 and 1987, Carlton Duty, who farms land near the domestic WWTF, discharged tile drain water to Pond 3. The Discharger collected samples from Pond 3 for selenium that revealed concentrations ranging from 11 to 32 µg/L between March 1985 and June 1989. In 1991 the Discharger collected samples for selenium from all its Ponds, which revealed low results in Pond A, B, 1 and 2 (typically < 2 µg/L), but elevated concentrations in Pond 3 (i.e., between 4 and 6 µg/L). Due to the elevated concentrations of selenium found in Pond 3, the California

Department of Fish & Game (DFG) recommended that the Discharger sample invertebrates for selenium. While the Discharger never sampled invertebrates for selenium, it did continue to monitor ponds for selenium. Recent monitoring by the Discharger did not show detectable levels of selenium until October 2000. Around this time, the Discharger changed laboratories and began construction activities that required significant dewatering. Self-monitoring data for 2001 indicate very high selenium concentrations in Ponds 2 and 3 (i.e., above 200 µg/L).

26. By letter dated 30 August 2001, the Discharger provided average concentrations from four background wells identified in Attachment D. The results from 22 analyses indicate the following concentrations in mg/L: aluminum of 2.3, calcium of 320, iron of 2.6, magnesium of 440, manganese of 0.06, selenium of 0.08, and sodium of 2,900.
27. The EPA and Regional Board provide criteria to apply to a potential water source before that water source can be considered unsuitable for municipal or domestic beneficial uses. If the TDS concentration in groundwater exceeds 10,000 mg/L (EPA) or 3,000 (Regional Board), groundwater may be considered unsuitable as a drinking water source. The Report indicates that between 1991 and 2000, LNAS collected 550 samples of groundwater for TDS from the shallow-upper aquifer for TDS. More than 97 percent of the samples exceeded the State criterion of 3,000 mg/L, of which more than 70 percent exceeded the federal criterion of 10,000 mg/L. As the TDS levels in the shallow-upper aquifer will also prohibit plant growth, the Report concludes that the shallow-upper aquifer cannot potentially be used for municipal, domestic, or agricultural supply.
28. A water balance was performed to quantify the amount of effluent that might be percolating from the domestic WWTF's two stabilization and three evaporation ponds. The average wastewater inflow to the domestic WWTF is about 1.9 mgd (2,130 acre-feet/year). Rainfall near LNAS totals about 8 inches/year, which equates to an annual inflow to the ponds of about 230 acre-feet. To determine the amount of evaporation from the ponds, an evaporation rate of 4.91 feet/year was used, which corresponds to 1,670 acre-feet/year of wastewater/effluent/rainfall evaporating from the ponds. As such, the remaining wastewater/effluent/rainfall (690 acre-feet/year) will percolate to groundwater (assuming the amount of water in ponds remains the same). The 690 acre-feet/year of percolate corresponds with a permeability of 2.0×10^{-6} cm/s, which suggests that the permeability of soils is higher than that given in Finding No. 7.
29. The domestic WWTF appears to be within the 50-year floodplain.
30. In dry weather, the Discharger routes water collected in the base's storm water collection system (e.g., landscape irrigation runoff) to a sump and then to Pond B. In wet weather, the Discharger routes storm water to the Kings River under the terms and conditions of State Water Resources Control Board Water Quality Order No. 97-03-DWQ, National Pollutant Discharge Elimination System General Permit No. CAS000001.
31. Land use in the WWTF vicinity and base operations primarily consists of agriculture. According to Land Use Maps prepared by DWR in 1996, the predominant crop in the WWTF vicinity is cotton. Other crops grown near LNAS include corn, alfalfa, safflower, and pistachios. Crops are predominantly irrigated with surface water.

32. Currently, the Discharger does not recycle wastewater from the domestic WWTF. The Discharger documented its efforts to recycle effluent from its domestic WWTF in *Final Water Reuse Management Study* (Reuse Study), dated August 2001, and prepared by Tetra Tech, Inc. The Reuse Study investigated the feasibility of recycling wastewater treated to standards for unrestricted reuse (i.e., disinfected tertiary treated wastewater). The Reuse Study provided the logistics for the Discharger to upgrade its treatment process and recycle disinfected tertiary treated wastewater on (a) a proposed golf course or (b) landscaping and/or agriculture.
33. Guidelines for identifying the quality of irrigation water necessary to sustain various crops were compiled by Ayers and Westcot in 1985 (Food and Agriculture Organization of the United Nations Irrigation Drainage Paper No. 29). The Basin Plan recognizes these Guidelines for providing relevant numerical criteria to evaluate compliance with the previously described narrative water quality objectives. The Guidelines are intended for use in estimating the potential hazards to crop production associated with long term use of the particular water being evaluated. According to the Guidelines, reductions in crop yields are not evident when irrigating (a) corn with water having an EC of less than 1,100 $\mu\text{mhos/cm}$, and (b) cotton with water having an EC less than 5,100 $\mu\text{mhos/cm}$. The Guidelines also indicate that a reduction in corn yields may occur when soil water extract contains boron ranging from 2.0 to 4.0 mg/L and cotton yields when the soil water extract contains boron ranging from 6.0 to 15.0 mg/L. The Guidelines explain that the maximum concentrations of boron in irrigation water where reductions in crop yields may occur will be slightly less. Using the relationship between soil water extract and irrigation water established in the Guidelines for EC would indicate that reductions in corn yields may occur when irrigation water contains boron ranging from 1.3 to 2.6 mg/L and cotton yields when irrigation water contains boron ranging from 4 to 10 mg/L.

Basin Plan and Regulatory Considerations

34. The *Water Quality Control Plan for the Tulare Lake Basin, Second Edition*, (hereafter Basin Plan) designates beneficial uses, establishes water quality objectives, contains implementation plans and policies for protecting waters of the Basin, and incorporates by references plans and policies adopted by the State Water Resources Control Board. These requirements implement the Basin Plan.
35. State Water Resources Control Board (SWRCB) Resolution No. 68-16 (hereafter Resolution 68-16 or the "Antidegradation" Policy) requires the Regional Board in regulating the discharge of waste to maintain high quality waters of the State until it is demonstrated that any change in quality will be consistent with maximum benefit to the people of the State, will not unreasonably affect beneficial uses, and will not result in water quality less than that described in the Regional Board's policies (e.g., quality that exceeds water quality objectives).
36. The Basin Plan identifies existing and potential beneficial uses of the Kings River from Peoples Weir to Stinson Weir on the North Fork and to Empire Weir No. 2 on the South Fork as agricultural, water contact recreation, noncontact water recreation, warm freshwater habitat, wildlife habitat, and groundwater recharge.
37. The Basin Plan identifies existing and potential beneficial uses of area groundwater as municipal and domestic supply, agricultural supply, industrial service supply, and industrial process supply.

38. The Basin Plan identifies the greatest long-term problem facing the entire Tulare Lake Basin as the increase in salinity in groundwater, which has accelerated due to intensive use of soil and water resources by irrigated agriculture. The Basin Plan recognizes that degradation is unavoidable until a valley wide drain is constructed to carry salt out of the basin. Until the drain is available, the Basin Plan describes numerous salt management recommendations and requirements.
39. The Basin Plan establishes requirements for wastewater treatment facilities that discharge to land in a manner that waste may infiltrate below the ground surface and degrade groundwater. These include a requirement that treatment systems with a design flow greater than 1 mgd provide removal of 80 percent or reduction to 40 mg/L, whichever is more restrictive, of both five-day biochemical oxygen demand and suspended solids. Another is a requirement that discharges to land from wastewater treatment facilities that have the potential to degrade groundwater not have an EC greater than source water plus 500 μ mhos/cm.
40. The shallow-upper aquifer is not of high quality and not useable or used for municipal and domestic supply, or agricultural supply. The intermediate-upper aquifer is also not of high quality and not used, and possibly not useable, for domestic or agricultural supply. Hydraulic continuity between aquifers is restricted. The poor quality of groundwater in the shallow-upper aquifer has been and continues to be affected by uncontrollable sources of waste, and therefore attainment of water quality objectives is not reasonable. High quality water for all uses is extracted from below the E-clay or obtained from the California Aqueduct.
41. As the groundwater affected by the discharge is not of high quality for salinity and nutrient waste constituents typically in municipal wastewater, Resolution 68-16 does not apply to the domestic WWTF and implementation of best practicable treatment and control is not required for the domestic WWTF. As long as waste constituents typically in municipal wastewater do not degrade the affected groundwater further, technology-based standards in the Basin Plan do not directly apply to the domestic WWTF. However, technology must be sufficient to assure consistency with other policies (e.g., Regional Board's recycling policy) and to remove waste constituents not acceptable for discharge into the groundwater. Until such time as the Discharger implements water recycling, it is required, at a minimum, to conduct the domestic WWTF discharge in a manner that does not cause or threaten to cause (a) nuisance conditions or (b) adverse effects on wildlife. As the groundwater affected by the discharge is not degraded for constituents removed by the industrial WWTF, Resolution 68-16 applies to the industrial WWTF and its discharge for these constituents (e.g., cadmium, dichloromethane, nickel) and implementation of best practicable treatment and control is required for the industrial WWTF discharge to the domestic WWTF.
42. As explained in Finding Nos. 11 and 13, the Discharger's domestic WWTF receives a significant portion of tile drain water, which could result in degradation for waste constituents typically not encountered in municipal wastewater. But as the Discharger disposes of the tile drain water in a manner that is consistent with the Basin Plan (i.e., evaporation ponds) and underlying groundwater is of comparable mineral quality; the discharge does not have a reasonable potential to degrade groundwater for salinity constituents, and therefore, the discharge of salinity waste constituents is consistent with Resolution 68-16.
43. Water in the Tulare Lake Basin is in short supply, requiring importation of surface waters from other parts of the State. The Basin Plan encourages recycling on irrigated crops wherever feasible

and indicates that evaporation of recyclable wastewater is not an acceptable permanent disposal method where the opportunity exists to replace an existing use or proposed use of fresh water with recycled water.

44. The Discharger indicates it is too expensive for it to recycle treated wastewater, as proposed in the Reuse Study (described in Finding No. 32). However, the Reuse Study only considered recycling options in which the Discharger upgrades its treatment. As the Discharger could recycle effluent on cotton (a salt tolerant crop that will not require the Discharger to upgrade its treatment facility), it is appropriate for the Discharger to submit for Regional Board consideration a recycling feasibility study that considers this and similar options.
45. While the discharge or infiltration of tile drain water into the Discharger's treatment system may not interfere with the Discharger's ability to comply with this Order's discharge specifications, it will render effluent unsuitable for recycling. This is inconsistent with the Regional Board's recycling policies (see Finding No. 43). Accordingly, it is appropriate for the Discharger to evaluate the possibility of (a) preventing tile drain water from entering its treatment system and (b) recycling treated wastewater on fodder crops (e.g., cotton) grown in the vicinity of its ponds.
46. As indicated in Finding No. 13, it appears that about 1.05 mgd (1,180 acre-feet/year) of wastewater originates from domestic sources. The water requirements for cotton near LNAS should be about 3.8 feet/year. If the Discharger were able to remove the tile drain component from its discharge, it would be able to provide enough effluent to grow about 310 acres of cotton per year. This would save about 5,900 acre-feet of fresh water in a five-year period.
47. The Basin Plan indicates that studies have shown ponds that receive tile drain water have the potential to impact wildlife and birds due to salinity and avian disease. To prevent such impacts, the Basin Plan recommends a number of requirements. Those that are relevant to this discharge include:
 - a. Removing attractive habitat, such as vegetation; and
 - b. Implementing a program for avian and waterfowl disease prevention, surveillance and control.
48. The Basin Plan also indicates that waterfowl may experience reduced hatchability and teratogenic impacts if evaporation ponds contain concentrations of selenium greater than 2.7 µg/L. The U. S. Fish and Wildlife Service (USFWS) established a wildlife impact selenium threshold level of 2 µg/L. As water quality monitoring indicates that selenium is well above these impact levels, it is appropriate for the Discharger to conduct hazing and perform egg monitoring to determine the extent of the contamination. If selenium contamination exists (egg selenium results exceed 6 mg/kg), it is appropriate that USFWS protocols be used to determine the amount and type of wildlife habitat necessary to mitigate wildlife impacts.
49. The discharge authorized herein and the treatment and storage facilities associated with the discharge, except for discharges of residual sludge and solid waste, are exempt from the requirements of Title 27, CCR, section 20005 et seq. (hereafter Title 27). The exemption, is pursuant to Title 27 section 20090 (a) and (b).

50. Section 13267 of the California Water Code (CWC) states, in part, that:

“In conducting an investigation specified in [section 13267] subdivision (a), the regional board may require that any person who has discharged, discharges, or is suspected of having discharged or discharging, or who proposes to discharge waste within its region, or any citizen or domiciliary, or political agency or entity of this state who has discharged, discharges, or is suspected of having discharged or discharging, or who proposes to discharge, waste outside of its region that could affect the quality of waters within its region shall furnish, under penalty of perjury, technical or monitoring program reports which the regional board requires. The burden, including costs, of these reports shall bear a reasonable relationship to the need for the report and the benefits to be obtained from the reports. In requiring those reports, the regional board shall provide the person with a written explanation with regard to the need for the reports, and shall identify the evidence that supports requiring that person to provide the reports.”

51. The monitoring and reporting program required by this Order and the attached Monitoring and Reporting Program No. R5-2002-0062 are necessary to assure compliance with these waste discharge requirements. The Discharger operates the facility that discharges the waste subject to this Order.
52. The action to revise waste discharge requirements for this existing facility is exempt from the provisions of the California Environmental Quality Act (CEQA) in accordance with Title 14, CCR, section 15301.
53. Pursuant to CWC section 13263(g), discharge is a privilege, not a right, and adoption of this Order does not create a vested right to continue the discharge.
54. The Regional Board considered all the above and the supplemental information and details in the attached Information Sheet, which is incorporated by reference herein, in establishing the following conditions of discharge.
55. The Regional Board notified the Discharger and interested agencies and persons of its intent to prescribe waste discharge requirements for this discharge and has provided them with an opportunity for a public hearing and an opportunity to submit their written views and recommendations.
56. The Regional Board, in a public meeting, heard and considered all comments pertaining to the discharge.

IT IS HEREBY ORDERED that Waste Discharge Requirements Order No. 86-173 is rescinded and that, pursuant to CWC sections 13263 and 13267, the U.S. Department of Navy Lemoore Naval Air Station, its agents, successors, and assigns, in order to meet the provisions contained in Division 7 of the CWC and regulations adopted thereunder, shall comply with the following:

[Note: Other prohibitions, conditions, definitions, and some methods of determining compliance are contained in the attached "Standard Provisions and Reporting Requirements for Waste Discharge Requirements" dated 1 March 1991.]

A. Discharge Prohibitions

1. Discharge of wastes to surface waters or surface water drainage courses is prohibited.

2. Bypass or overflow of untreated or partially-treated waste is prohibited, except as allowed in Provision E.2 of Standard Provisions and Reporting Requirements.
3. Discharge of waste classified as 'hazardous,' as defined in section 2521(a) of Title 23, CCR, section 2510 et seq., or 'designated,' as defined in CWC section 13173, is prohibited.
4. Recycling of effluent to areas without Regional Board-adopted water recycling requirements or waiver of said requirements is prohibited.

B. Industrial WWTF Discharge Specifications

1. The monthly average daily discharge flow shall not exceed 0.05 mgd.
2. The discharge shall not exceed the following limits:

<u>Constituent</u>	<u>Daily Maximum</u>	<u>Monthly Average</u>
TSS (mg/L)	60	30
Settleable Solids (mL/L)	0.5	0.1

3. The discharge shall not exceed the following limits in (mg/L):

<u>Constituent</u>	<u>Value¹</u>
Copper	3.4
Nickel	0.1
Total Chromium	0.5
Zinc	2.6
Total Cyanide	1.2
Silver	0.4
Lead	0.3
Cadmium	0.1
Dichloromethane	0.105
Trichloroethane ²	0.105

¹ Calculated by a 3-month moving average

² 1,1,1-Trichloroethane plus 1,1,2-Trichloroethane

4. The oil and grease concentration of the discharge shall not exceed 100 mg/L, as calculated by a 12-month moving average.

C. Domestic WWTF Discharge Specifications

1. The monthly average daily discharge flow shall not exceed 2.12 mgd.
2. The pH of the discharge shall not be less than 6.0 or greater than 9.0.

3. Objectionable odors originating at the WWTF shall not be perceivable beyond the limits of the waste treatment areas and effluent disposal ponds at an intensity that creates or threatens to create nuisance conditions.
4. As a means of discerning compliance with Discharge Specification C.3, the dissolved oxygen content in the upper zone (one foot) of wastewater in all ponds shall not be less than 1.0 mg/L.
5. Ponds shall be managed to prevent breeding of mosquitoes and impacts to wildlife. In particular:
 - a. An erosion control plan should assure that small coves and irregularities are not created around the perimeter of the water surface.
 - b. Weeds shall be minimized through control of water depth, harvesting, and herbicides.
 - c. Dead algae, vegetation, and debris shall not accumulate on the water surface.
 - d. Vegetation management operations in areas in which nesting birds have been observed shall be carried out either before or after, but not during, the **April 1 to June 30** bird nesting season.
6. Freeboard shall never be less than two feet in any pond (measured vertically).
7. As a means of discerning compliance with Domestic WWTF Discharge Specifications C.6 and C.9, the Discharger shall install and maintain in each pond permanent markers with calibration indicating the water level at design capacity and available operational freeboard. Upon the Discharger's written request, specific WWTF ponds may be exempt from this requirement. Such exemptions shall be subject to the Executive Officer's written approval.
8. The Discharger shall, in consultation with USFW, implement mitigation measures appropriate to this discharge. This may include the conduct of a hazing program during the months of February through July that effectively keeps birds moving. If necessary, hazing shall, at a minimum, be conducted at each pond where birds are present and where the geometric mean (i.e., the nth root of a product of n factors) selenium concentration in water is equal to or greater than 2.0 µg/L. If a nest containing one or more eggs is established despite mitigation measures taken, hazing shall be terminated in proximity of the nest for the remainder of the season unless the Discharger determines in consultation with USFWS that hazing should continue. The Discharger and USFWS will determine the effectiveness of continued hazing under circumstances where a nest has been established and report the results of such hazing to the Regional Board. The Discharger shall comply with this specification until it receives written notification from the Executive Officer that, due to changed site conditions, hazing is no longer necessary.
9. The Discharger shall manage ponds to maintain a minimum depth of at least 2.0 feet or more if necessary to control vegetation.
10. When draining or filling a pond, the Discharger shall employ all feasible measures to attain zero depth or the required minimum depth as quickly as possible.

11. The WWTF shall be designed, constructed, operated, and maintained to prevent inundation or washout due to floods with a 100-year frequency.
12. The Discharger shall preclude public access to the waste treatment and effluent disposal facilities through methods such as fences, signs, or other acceptable means.
13. Ponds shall have sufficient capacity to accommodate allowable wastewater flow and design seasonal precipitation and ancillary inflow and infiltration during the winter. Design seasonal precipitation shall be based on total annual precipitation using a return period of 100 years, distributed monthly in accordance with historical rainfall patterns.
14. On **15 November** of each year, available storage capacity in ponds shall be at least equal to the volume necessary to comply with Domestic WWTF Discharge Specification C.13.

D. Sludge Specifications

Sludge in this document means the solid, semisolid, and liquid residues removed during primary, secondary, or advanced wastewater treatment processes. Solid waste refers to grit and screening material generated during preliminary treatment. Residual sludge means sludge that will not be subject to further treatment at the industrial and domestic WWTFs. Biosolids refers to domestic WWTF sludge that has undergone sufficient treatment and testing to qualify for reuse pursuant to federal and state regulations as a soil amendment for agriculture, silviculture, horticulture, and land recycling.

1. Sludge and solid waste shall be removed from the industrial and domestic WWTFs, as needed to ensure optimal plant operation.
2. Treatment and storage of sludge generated by each WWTF shall be confined to the respective WWTF property and conducted in a manner that precludes infiltration of waste constituents into soils.
3. Any storage of residual sludge, solid waste, and biosolids on property of each WWTF shall be temporary and controlled and contained in a manner that minimizes leachate formation.
4. Residual sludge, biosolids, and solid waste shall be disposed of in a manner approved by the Executive Officer and consistent with Title 27. Removal for further treatment, disposal, or reuse at sites (i.e., landfill, WWTF, composting sites, soil amendment sites) operated in accordance with valid waste discharge requirements issued by a regional water quality control board will satisfy this specification.
5. Use of biosolids as a soil amendment shall comply with valid waste discharge requirements issued by a regional water quality control board. In most cases, this will mean General Biosolids Order (SWRCB Water Quality Order No. 2000-10-DWQ, *General Waste Discharge Requirements for the Discharge of Biosolids to Land for Use as a Soil Amendment in Agricultural, Silvicultural, Horticultural, and Land Reclamation Activities*). For a biosolids use project to be covered by the General Biosolids Order, the Discharger must file a complete Notice of Intent and receive a Notice of Applicability for each project.

6. Use and disposal of biosolids should comply with the self-implementing federal regulations of Title 40, Code of Federal Regulations (CFR), Part 503, which are subject to enforcement by the U. S. Environmental Protection Agency (EPA), not the Regional Board. If during the life of this Order the State accepts primacy for implementation of 40 CFR 503, the Regional Board may also initiate enforcement where appropriate.

E. Provisions

1. The Discharger shall comply with *Standard Provisions and Reporting Requirements for Waste Discharge Requirements*, dated 1 March 1991, which are attached hereto and by reference a part of this Order. This attachment and its individual paragraphs are commonly referenced as *Standard Provision(s)*.
2. The Discharger shall comply with Monitoring and Reporting Program (MRP) No. R5-2002-0062, which is part of this Order, and any revisions thereto as ordered by the Executive Officer.
3. A copy of this Order, including its attachments and Standard Provisions, shall be kept at the WWTF for reference by operating personnel. Key operating personnel shall be familiar with its contents.
4. The Discharger shall use best practicable treatment and control, including proper operation and maintenance, to comply with terms of this Order pertaining to the industrial WWTF and its discharge to the domestic WWTF.
5. All technical reports required herein that involve planning, investigation, evaluation, or design, or other work requiring interpretation and proper application of engineering or geologic sciences, shall be prepared by or under the direction of persons registered to practice in California pursuant to California Business and Professions Code sections 6735, 7835, and 7835.1. To demonstrate compliance with sections 415 and 3065 of Title 16, CCR, all technical reports must contain a statement of the qualifications of the responsible registered professional(s). As required by these laws, completed technical reports must bear the signature(s) and seal(s) of the registered professional(s) in a manner such that all work can be clearly attributed to the professional responsible for the work.
6. By **1 August 2002** of the adoption of this Order, the Discharger shall submit a technical report that contains a certification that its domestic WWTF is designed, constructed, operated, and maintained to prevent inundation or washout due to floods with a 100-year frequency. The technical report shall comply with Provision E.5 and is subject to Executive Officer approval.
7. By **1 August 2002** of the adoption of this Order, the Discharger shall submit a report that details the measures it will implement to determine potential sources of dichloromethane to its industrial WWTF. The report is subject to Executive Officer approval.
8. By **1 February 2003**, the Discharger shall submit a technical report that describes the effect of tile drain water on the efficacy of its treatment system. The technical report shall also propose a time schedule for evaluating the integrity of the Discharger's sewer collection system. The technical report shall comply with Provision E.5 and is subject to Executive Officer approval.

9. By **1 September 2002** of the adoption of this Order, the Discharger shall submit a report for Executive Officer approval outlining EPA test methods and detection limits for priority pollutants listed in Title 40, Code of Federal Regulations, Part 131.
10. By **1 September 2002** of the adoption of this Order, and every three years thereafter, the Discharger shall submit a written technical report prepared by a qualified wildlife biologist assessing whether mitigation measures have been fully implemented and whether the measures, as implemented, fully compensate for all existing and potential impacts on the target species. The report shall be sufficiently comprehensive and statistically sound to determine whether complete mitigation has been, and can continue to be achieved. The Regional Board, after review of this report, may amend this Order to prohibit further discharge or modify mitigation measures.
11. By **1 November 2002** of the adoption of this Order, the Discharger shall submit a proposed time schedule for repairing its leachate collection system.
12. By **1 May 2003** of the adoption of this Order, the Discharger shall submit a feasibility study that describes opportunities to recycle effluent on agricultural lands where fresh water is currently used. The study shall also include implementation schedules for each identified opportunity. The feasibility study shall comply with Provision E.5 and is subject to Executive Officer approval.
13. The Discharger shall not allow pollutant-free wastewater to be discharged into the WWTF collection, treatment, and disposal systems in amounts that significantly diminish the system's capability to comply with this Order.
14. The Discharger shall dispose of bird carcasses as directed by DFG. Upon finding an unusual number (i.e., more than 15), the Discharger shall notify DFG within 24 hours and a bird carcass shall, at the DFG's discretion, be held for diagnosis. The Discharger shall keep a record of the number, date, and species of carcasses found and burned or buried.
15. Subject to prior notice to, and contingent upon a written report of findings being submitted to, the Regional Board, Standard Provision A.8 shall include employees of DFG, USFWS, and U. S. Geological Survey – Biological Resources Division to the extent necessary to monitor compliance with mitigation measures to prevent wildlife impacts that are specified in this Order.
16. Operations of the Discharger's domestic WWTF ponds shall not violate the Migratory Bird Treaty Act.
17. The Discharger shall report to the Regional Board any toxic chemical release data it reports to the State Emergency Response Commission within 15 days of reporting the data to the Commission pursuant to section 313 of the "Emergency Planning and Community Right to Know Act of 1986." If the Regional Board determines that the toxic waste constituent had or has a reasonable potential to cause or contribute to violation of a water quality objective, the Regional Board may reopen this Order and prescribe an effluent limitation for the constituent.
18. The Discharger must comply with all conditions of this Order, including timely submittal of technical and monitoring reports as directed by the Executive Officer. Accordingly, the

Discharger shall submit to the Regional Board on or before each report due date the specified document or, if an action is specified, a written report detailing evidence of compliance with the date and task. If noncompliance is being reported, the reasons for such noncompliance shall be stated, plus an estimate of the date when the Discharger will be in compliance. The Discharger shall notify the Regional Board by letter when it returns to compliance with the time schedule. Violations may result in enforcement action, including Regional Board or court orders requiring corrective action or imposing civil monetary liability, or in revision or rescission of this Order.

19. In the event of any change in control or ownership of land or waste treatment and storage facilities presently owned or controlled by the Discharger, the Discharger shall notify the succeeding owner or operator of the existence of this Order by letter, a copy of which shall be immediately forwarded to this office. To assume operation under this Order, the succeeding owner or operator must apply in writing to the Executive Officer requesting transfer of the Order. The request must contain the requesting entity's full legal name, the state of incorporation if a corporation, the address and telephone number of the persons responsible for contact with the Regional Board and a statement. The statement shall comply with the signatory paragraph of Standard Provision B.3 and state that the new owner or operator assumes full responsibility for compliance with this Order. Failure to submit the request shall be considered a discharge without requirements, a violation of the California Water Code. Transfer shall be approved or disapproved in writing by the Executive Officer.
20. The Regional Board will review this Order periodically and will revise requirements when necessary.

I, GARY M. CARLTON, Executive Officer, do hereby certify the foregoing is a full, true, and correct copy of an Order adopted by the California Regional Water Quality Control Board, Central Valley Region, on 26 April 2002.


GARY M. CARLTON, Executive Officer

Order Attachments:

Monitoring Standard Provisions

A. Location Map

B. Industrial WWTF Flow Schematic

C. Domestic WWTF Flow Schematic

D. Background Monitoring Well Locations

Information Sheet

Standard Provisions (1 March 1991 version) (separate attachment to Discharger only)

RAS/JLK:fmc: 4/26/2002

CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL VALLEY REGION

MONITORING AND REPORTING PROGRAM NO. R5-2002-0062

FOR
U.S. DEPARTMENT OF THE NAVY
LEMOORE NAVAL AIR STATION
WASTEWATER TREATMENT FACILITY
KINGS COUNTY

This Monitoring and Reporting Program (MRP) is required pursuant to California Water Code section 13267. The Discharger shall not implement any changes to this MRP unless a revised MRP is adopted by the Regional Board or issued by the Executive Officer. All samples should be representative of the volume and nature of the discharge or matrix of material sampled. The time, date, and location of each sample shall be recorded on the sample chain of custody form. Attachments B and C indicate the sampling locations. All analyses shall be performed in accordance with Standard Provisions, Provisions for Monitoring.

**INDUSTRIAL WASTEWATER TREATMENT FACILITY
EFFLUENT MONITORING**

The Discharger shall collect effluent samples from the industrial wastewater treatment facility at a point following the final treatment unit, but before it combines with domestic sewage. Effluent monitoring shall include the following:

<u>Constituent</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling Frequency</u>
Flow	mgd	Metered	Continuous
Average Daily Flow	mgd	Calculated	Daily
pH	pH units	Grab	Weekly
EC	µmhos/cm	Grab	Weekly
Settleable Solids	mL/L	Grab	Weekly
Total Suspended Solids	mg/L	Grab	Weekly
Oil and Grease	mg/L	Grab	Twice/Monthly ¹
Dichloromethane	µg/L	Grab	Twice/Monthly ^{1,2}
Copper	mg/L	8-hr composite	Monthly
Nickel	mg/L	8-hr composite	Monthly
Total Chromium	mg/L	8-hr composite	Monthly
Zinc	mg/L	8-hr composite	Monthly
Silver	mg/L	8-hr composite	Monthly
Lead	mg/L	8-hr composite	Monthly
Cadmium	mg/L	8-hr composite	Monthly
Total Cyanide	mg/L	Grab	Monthly
Trichloroethane ³	µg/L	Grab	Monthly
Priority Pollutants	µg/L	Grab	Annually ⁴

¹ Nonconsecutive weeks

² After one year, frequency may be reduced to monthly

³ 1,1,1-Trichloroethane and 1,1,2-Trichloroethane

⁴ October

DOMESTIC WASTEWATER TREATMENT FACILITY INFLUENT MONITORING

The Discharger shall collect influent samples at the headworks prior to any treatment of waste. Influent monitoring shall include at least the following:

<u>Constituent</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling Frequency</u>
Maximum Daily Flow	mgd	Continuous	Continuous
Average Daily Flow	mgd	Computed	Daily ¹
Monthly Average Flow	mgd	Computed	Monthly
Settleable Solids	mL/L	Grab	Weekly
pH	pH units	Grab	Weekly
BOD ₅ ²	mg/L	8-hr Composite ³	Monthly ⁴
TSS ⁵	mg/L	8-hr Composite ³	Monthly ⁴

¹ Sample frequencies referenced hereafter in this program as daily shall not include weekends or holidays.

² Five-day, 20°C biochemical oxygen demand

³ A representative composite sample may be obtained by collecting four samples at uniform intervals over an eight-hour period.

⁴ Quarterly after one year (January, April, July, and October)

⁵ Total Suspended Solids

EFFLUENT MONITORING

The Discharger shall collect effluent samples at a point in the system following treatment and before discharge to evaporation Pond 1. Effluent samples shall be representative of the volume and nature of the discharge. Effluent monitoring shall include the following:

<u>Constituent</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling Frequency</u>
Settleable Solids	mL/L	Grab	Weekly
pH	pH Units	Grab	Weekly
EC ¹	µmhos/cm	Grab	Monthly
BOD ₅	mg/L	8-hr Composite	Monthly ²
TSS	mg/L	8-hr Composite	Monthly ²
Total Dissolved Solids (TDS) ³	mg/L	8-hr Composite	Monthly ^{2,4}
Nitrate (as N)	mg/L	8-hr Composite	Semiannually ⁵
Total Kjeldahl Nitrogen (TKN)	mg/L	8-hr Composite	Semiannually ⁵
Total Nitrogen	mg/L	Calculated	Semiannually ⁵
Antimony	µg/L	Grab	Annually ⁶

<u>Constituent</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling Frequency</u>
Molybdenum	µg/L	Grab	Annually ⁶
Arsenic	µg/L	Grab	Annually ⁶
Selenium	µg/L	Grab	Twice/Monthly ^{2,4}
General Minerals ⁷	mg/L	Grab	Monthly ^{2,4}

¹ Conductivity at 25°C

² After one year, frequency may be reduced to quarterly (i.e., January, April, July, and October).

³ TDS referenced hereafter in this program shall be determined using EPA Method No. 160.1 for combined organic and inorganic TDS and EPA Method No. 160.4 for inorganic TDS or equivalent analytical procedures specified in 40 CFR Part 136.

⁴ Shall coincide with EC sampling

⁵ April and October

⁶ October

⁷ General Minerals as referred to in this program shall include the constituents in the General Minerals Analyte List presented below. Using proper sampling methods and appropriate sample containers is critical in obtaining valid results for general minerals analyses. Follow laboratory directions and secure sample containers as appropriate for requesting analyses for general minerals.

General Minerals Analyte List

Alkalinity (as CaCO ₃)	Carbonate (as CaCO ₃)	Manganese
Aluminum	Chloride	Phosphate
Bicarbonate (as CaCO ₃)	Hardness (as CaCO ₃)	Potassium
Boron	Iron	Sodium
Calcium	Magnesium	Sulfate

POND MONITORING

The Discharger shall monitor each of its stabilization and evaporation ponds for the parameters specified below. Freeboard shall be monitored on all ponds in use to the nearest one tenth of a foot. Pond monitoring shall include at least the following:

<u>Constituent/Parameter</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling Frequency</u> ¹
Selenium	mg/L	Grab	Twice/Monthly
Dissolved Oxygen (DO)	mg/L	Grab ²	Weekly
Pond Depth	feet	Observation	Weekly
Freeboard	feet	Observation	Weekly

- ¹ If results of monitoring appear to violate discharge specifications, but monitoring frequency is not sufficient to validate violation or indicate a violation and potential upset of the treatment process (e.g., less than minimum dissolved oxygen concentration), the frequency of sampling shall be increased to confirm the magnitude and duration of violation, if any, and aid in identification and resolution of the problem.
- ² Samples shall be collected at a depth of one foot from each pond in use, opposite the inlet, and analyzed for DO. Samples shall be collected between 0700 and 0900 hours. Time of sampling shall be reported. If DO results for any pond in use indicate noncompliance with the discharge specification, the Discharger shall implement corrective measures as specified in the O&M manual and monitor said pond daily until its DO stabilizes above 1 mg/L.

In addition, the Discharger shall inspect the condition of the ponds once per week and write visual observations in a bound logbook. Notations shall include observations of whether weeds are developing in the water or along the bank, and their location; whether dead algae, vegetation, scum, or debris are accumulating on the pond surface and their location; whether burrowing animals or insects are present; and the color of the ponds (e.g., dark sparkling green, dull green, yellow, gray, tan, brown, etc.). A copy of the entries made in the log during each month shall be submitted along with the monitoring report the following month. If the Discharger finds itself in violation of Domestic WWTF Discharge Specifications C.3, C.4, C.5, C.6, C.9, or C.10 the Discharger shall briefly explain the action taken or to be taken to correct the violation. The Discharger shall certify in each November monitoring report that it is in compliance with Domestic WWTF Discharge Specification C.13.

SLUDGE MONITORING

Within 270 days of the adoption of this Order, the Discharger shall characterize sludge quality from its domestic treatment ponds for the following constituents:

Arsenic	Lead	Zinc	Potassium
Cadmium	Mercury	Sulfate	Sodium
Molybdenum	Nickel	Chloride	
Copper	Selenium	Boron	

To characterize sludge, the Discharger shall collect a composite sample from each of its ponds. For guidance in sampling, the Discharger should consult EPA's *POTW Sludge Sampling and Analysis Guidance Document*, August 1989. The Discharger shall submit the results of the sludge characterization to the Regional Board by the **1st day of the second month** following sampling.

WILDLIFE MONITORING

Wildlife monitoring at the domestic WWTF ponds shall be conducted by or under the direct supervision of a qualified wildlife biologist with or able to obtain a permit from the U. S. Fish and Wildlife Service to collect eggs.

Bird counts shall be conducted semimonthly throughout the year. If birds nest at the ponds, then breeding bird surveys shall be conducted semimonthly until completion (typically from March through

August and must include counts of nests, eggs and young. Nests shall be flagged, and five (5) random recurvirostrid eggs from five separate nests shall be collected and shall be chemically analyzed for total recoverable selenium and moisture content.

The Discharger shall inspect each domestic WWTF pond weekly for dead birds. Inspections shall be increased to daily at any pond when the water depth is less than two (2) feet and for all ponds when a botulism or fowl cholera outbreak is occurring in the area, as confirmed by the California Department of Fish & Game (DFG). Once DFG confirms the said outbreak is over, the Discharger may reduce the frequency of pond inspections back to weekly.

If the Discharger documents, after a minimum of one year of monitoring, that selenium levels in domestic WWTF ponds do not pose a threat to wildlife, it may submit a written request to the Executive Officer that wildlife monitoring be terminated. Should the Executive Officer concur that ponds do not pose a threat to wildlife, the Discharger will receive written notice that it may terminate wildlife monitoring.

REPORTING

The Discharger shall report monitoring data and information as required in this MRP and as required in the Standard Provisions and Reporting Requirements. All reports submitted in response to this MRP shall comply with the signatory requirements in Standard Provisions, General Reporting Requirements B.3. Daily, twice weekly, weekly, twice monthly, and monthly monitoring data shall be reported in monthly monitoring reports. Monthly monitoring reports shall be submitted to the Regional Board by the **1st day of second month following sampling**. Quarterly monitoring reports shall be submitted by **1st day of second month after the calendar quarter**.

In each monitoring report, the Discharger shall discuss the compliance record for the reporting period. If violations have occurred during the reporting period, the report shall also discuss the corrective actions taken and/or planned to bring the discharge into full compliance with this Order. Monitoring data and/or discussions submitted concerning WWTF performance must also be signed and certified by the chief plant operator. When reports contain laboratory analyses performed by the Discharger and the chief plant operator is not in the direct line of supervision of the laboratory, reports must also be signed and certified by the chief of the laboratory.

In reporting monitoring data, the Discharger shall arrange the data in tabular form so that the date, the constituents, and the concentrations are readily discernible. The data shall be summarized in such a manner that illustrates clearly whether the Discharger complies with waste discharge requirements. If the Discharger monitors any pollutant at the locations designated herein more frequently than is required by this Order, the results of such monitoring shall be included in the discharge monitoring report.

All wildlife monitoring shall be submitted to the Regional Board as follows:

<u>Reporting Period</u>	<u>Draft</u>	<u>Final</u>
October-March		1 May
<u>April-September</u>	20 December ¹	20 February

¹ Five copies for peer review

The Discharger may also be requested to submit an annual report to the Regional Board with tabular and graphical summaries of the monitoring data obtained during the previous year. Any such request shall be made in writing. The report shall discuss any corrective actions the Discharger takes or plans to take to bring the discharge into full compliance with the waste discharge requirements.

By **1 February of each year**, the Discharger shall submit a written report to the Executive Officer containing the following:

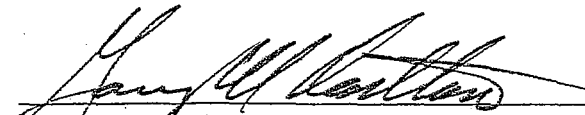
1. The names, certificate grades, and general responsibilities of all persons in charge of wastewater treatment and disposal.
2. The names and telephone numbers of persons to contact regarding the WWTF for emergency and routine situations.
3. A statement certifying when the flow meter and other monitoring instruments and devices were last calibrated, including identification of who performed the calibration (Standard Provision C.4). Reports submitted concerning WWTF performance must also be signed and certified by the chief plant operator. When reports contain laboratory analyses performed by the Discharger and the chief plant operator is not in the direct line of supervision of the laboratory, reports must also be signed and certified by the chief of the laboratory.
4. A statement whether the current operation and maintenance manual, and contingency plan, reflect the wastewater treatment facility as currently constructed and operated, and the dates when these documents were last reviewed for adequacy.
5. The results of an annual evaluation conducted pursuant to Standard Provision E.4 and a figure depicting monthly average discharge flow for the past five years.
6. The most recent LNAS annual water supply report.
7. A summary of sludge monitoring, including
 - a. Annual sludge production in dry tons and percent solids.
 - b. A schematic diagram showing sludge handling facilities and solids flow diagram.
 - c. A description of disposal methods, including the following information related to the disposal methods used at each WWTF. If more than one method is used, include the percentage of annual sludge production disposed of by each method.
 - i. For **landfill disposal**, include: (a) the Order numbers of WDRs that regulate the landfill(s) used, (b) the present classifications of the landfill(s) used, and (c) the names and locations of the facilities receiving sludge.
 - ii. For **land application**, include: (a) the locations of the site(s), and (b) the Order numbers of any WDRs that regulate the site(s).
 - iii. For **incineration**, include: (a) the names and location of the site(s) where sludge incineration occurs, (b) the Order numbers of WDRs that regulate the site(s), (c) the disposal method of ash, and (d) the names and locations of facilities receiving ash (if applicable).
 - iv. For **composting**, include: (a) the location of the site(s), and (b) the Order numbers of any WDRs that regulate the site(s).

MONITORING AND REPORTING PROGRAM NO. R5-2002-0062
U. S. DEPARTMENT OF NAVY LNAS WWTF
KINGS COUNTY

-7-

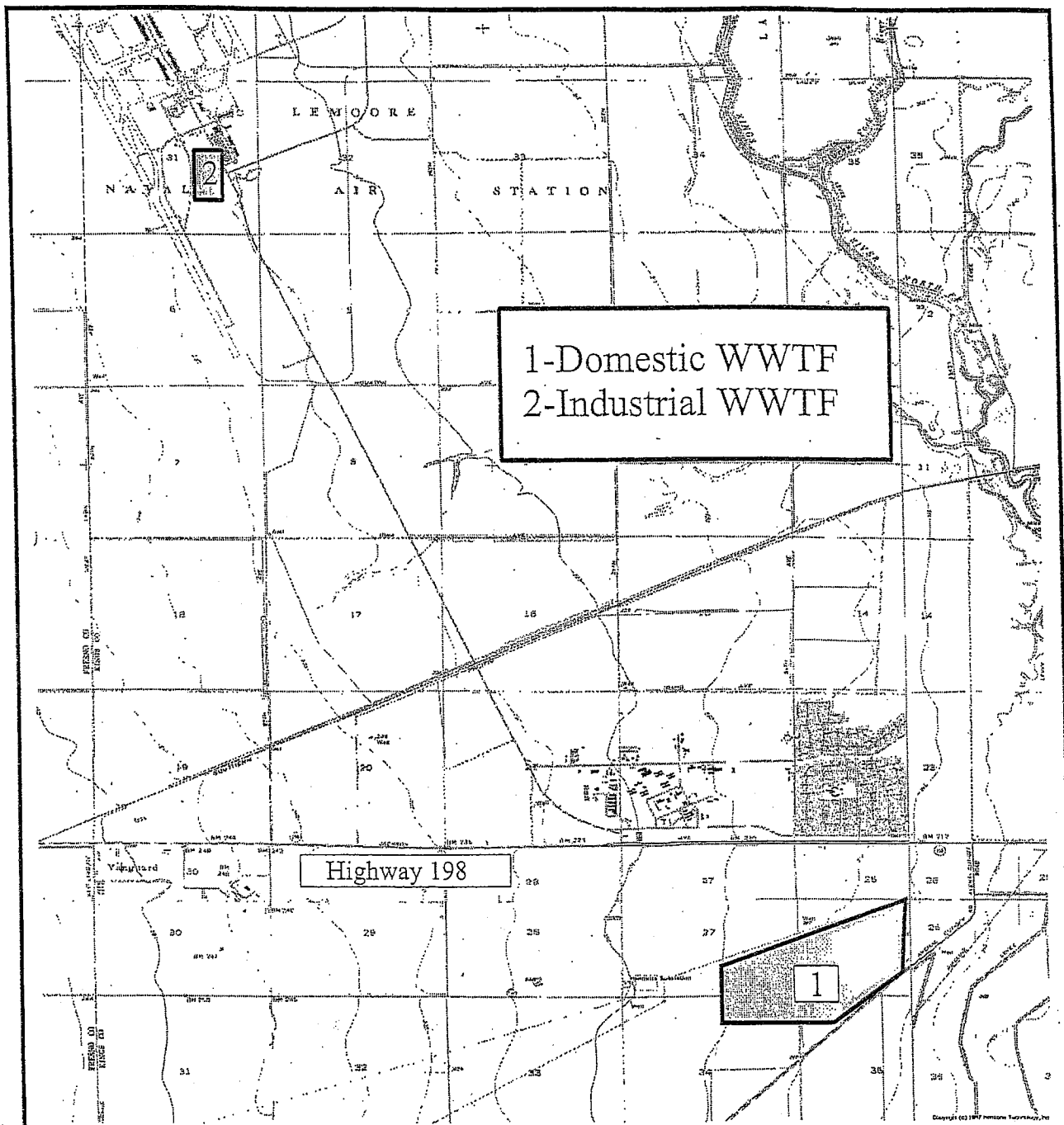
8. A summary and discussion of the measures taken to eliminate dichloromethane from the industrial WWTF discharge.
9. A summary and discussion of the compliance record for the reporting period. If violations have occurred, the report shall also discuss the corrective actions taken and planned to bring the discharge into full compliance with this Order.

The Discharger shall implement the above monitoring program on the first day of the month following adoption of this Order.


GARY M. CARLTON, Executive Officer

26 April 2002
(Date)

RAS/JLK:fmc: 4/26/2002

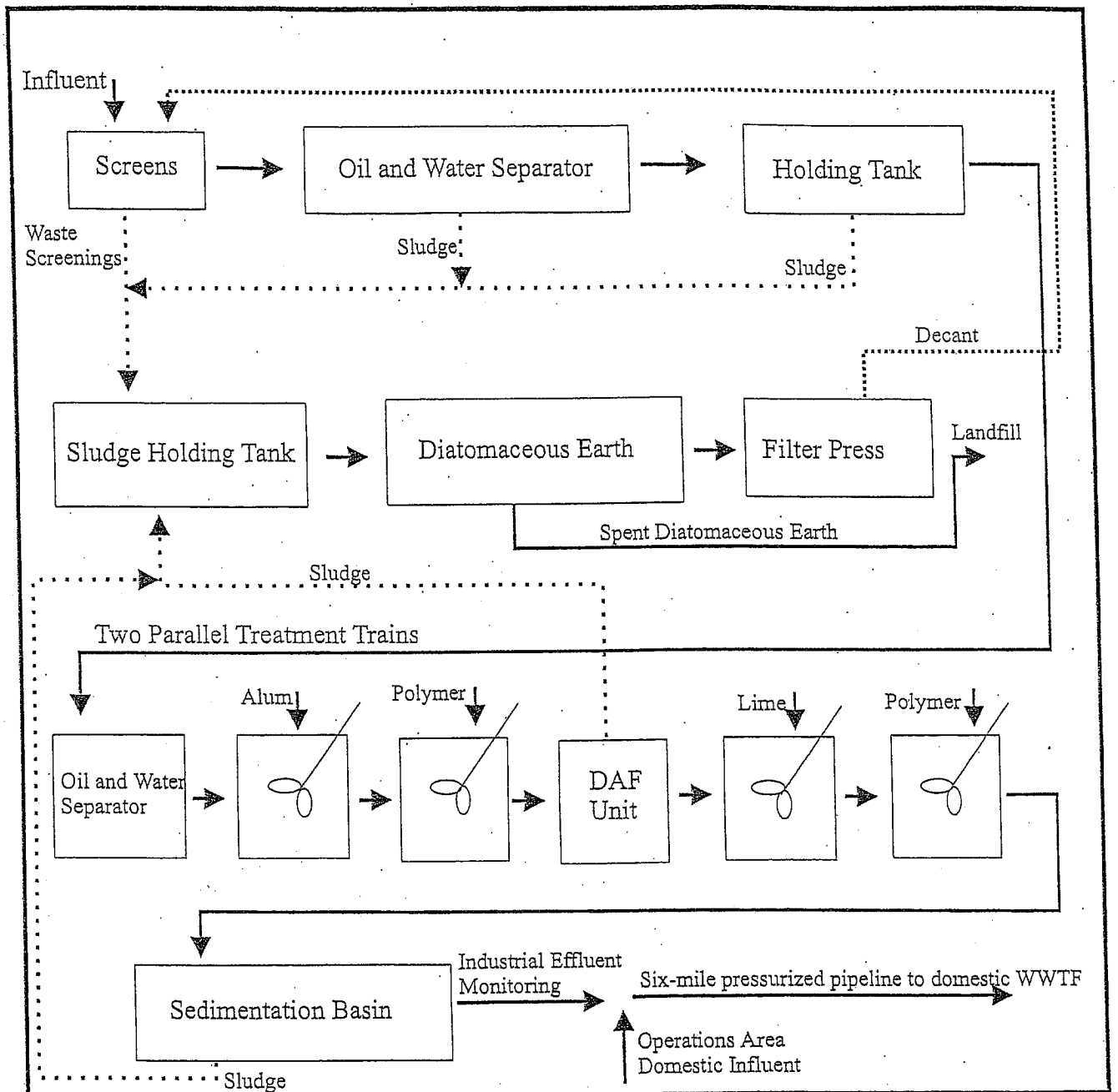


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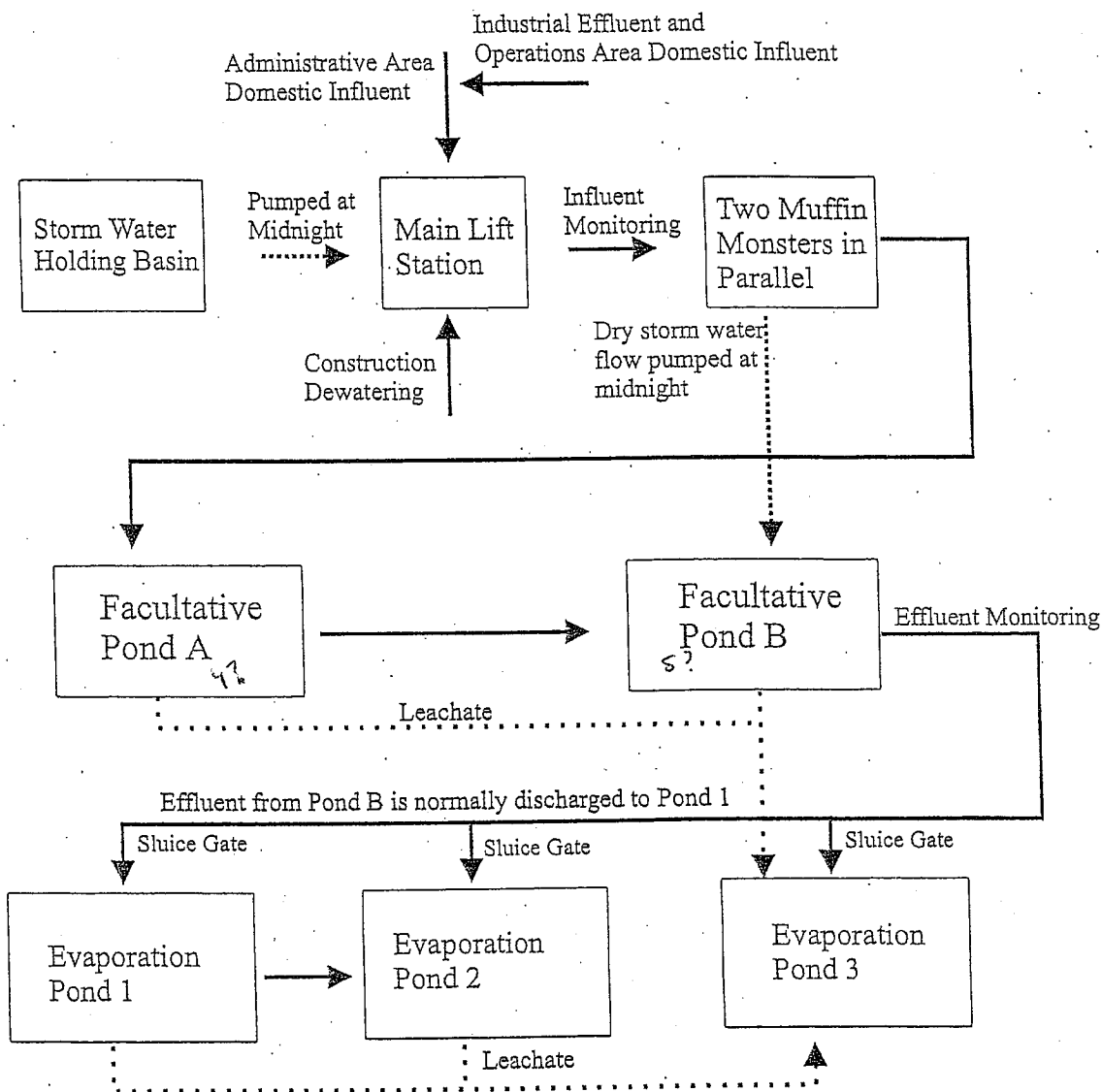
ATTACHMENT A
WASTE DISCHARGE REQUIREMENTS
ORDER NO. R5-2002-0062

LOCATION MAP
LEMOORE NAVAL AIR STATION
WASTEWATER TREATMENT FACILITY, KINGS COUNTY

Section 31, T18S and Sections 26, 27, 34, and 35, T19S, R19E, MDB&M
Vanguard and Westhaven 7.5 Min Quad Maps



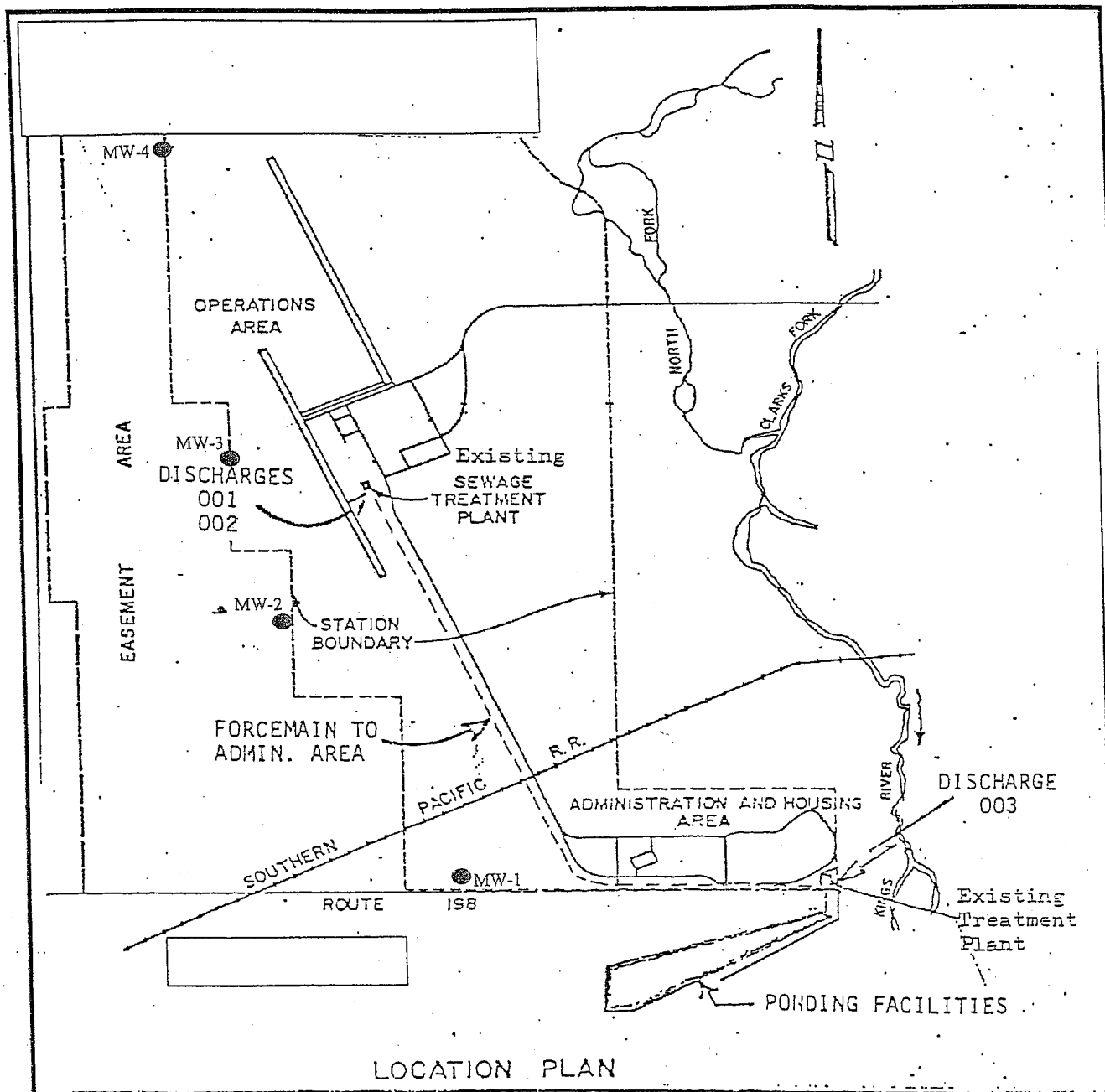
ATTACHMENT B
WASTE DISCHARGE REQUIREMENTS
ORDER NO. R5-2002-0062
INDUSTRIAL WWTf FLOW SCHEMATIC
LEMOORE NAVAL AIR STATION
WASTEWATER TREATMENT FACILITY, KINGS COUNTY



ATTACHMENT C
WASTE DISCHARGE REQUIREMENTS
ORDER NO. R5-2002-0062

DOMESTIC WWTF FLOW SCHEMATIC

LEMOORE NAVAL AIR STATION
WASTEWATER TREATMENT FACILITY, KINGS COUNTY



ATTACHMENT D
WASTE DISCHARGE REQUIREMENTS
ORDER NO. R5-2002-0062

BACKGROUND MONITORING WELL LOCATIONS

LEMOORE NAVAL AIR STATION
WASTEWATER TREATMENT FACILITY, KINGS COUNTY

INFORMATION SHEET

ORDER NO. R5-2002-0062

U. S. DEPARTMENT OF NAVY LEMOORE NAVAL AIR STATION
WASTEWATER TREATMENT FACILITY
KINGS COUNTY

Background

The U. S. Department of Navy, Lemoore Naval Air Station (LNAS or Discharger), operates industrial and domestic wastewater treatment facilities (WWTFs) about six miles west of the City of Lemoore in Kings County. About 90 percent of the wastewater discharged to the industrial WWTF consists of washdown from aircraft washing. The remaining wastewater discharged to the industrial WWTF includes washdown from jet engine test cells and bearings and brake shops. Domestic wastewater generated from station personnel and about 15,000 residents, industrial WWTF effluent, shallow groundwater from construction dewatering, and filter backwash water from the onsite water treatment plant are discharged to the domestic WWTF. Waste Discharge Requirements Order No. 86-173 prescribes terms and conditions for the industrial and domestic WWTFs. The industrial WWTF has a design capacity of 0.05 mgd and currently treats a monthly average flow of 0.014 mgd, while the domestic WWTF has a design capacity of 2.12 mgd and currently treats an average monthly flow of about 1.9 mgd. The Discharger disposes of wastewater by evaporation in ponds overlying low permeable soils.

History of the Wastewater Treatment Facilities

The LNAS was commissioned in 1961 to provide training operations for fleet jet aircraft crew personnel. Wastewater flow originates from an Operations Area and an Administration Area. The Operations Area generates about 0.02 mgd of industrial wastewater and about 0.2 mgd of domestic wastewater, while the Administration Area generates about 1.8 mgd of domestic wastewater. Until the Discharger completed a pipeline project in 1984, the Discharger treated these wastestreams in three separate WWTFs.

Between 1964 and 1974, the Discharger conveyed effluent from the Administration Area WWTF to the Kings River via a 0.75-mile unlined ditch. During this time, the Administrative Area WWTF consisted of trickling filters and sedimentation basins with a design flow of 2.0 mgd. The Discharger also discharged storm water to the unlined ditch and continues to do so during periods of storm runoff. In 1975, the Discharger installed the current pond system that consists of two stabilization ponds and three evaporation ponds that total 340 acres. The pond system eliminated the Discharger's need to discharge treated wastewater from the Administration Area to the Kings River.

Until 1984, the Discharger also used a trickling filter plant with a design flow of 0.3 mgd to treat domestic wastewater from its Operations Area. The Discharger treated industrial wastewater separately and disposed of effluent from the Operations Areas (i.e., both domestic and industrial wastewater) by irrigating fodder and fiber crops under the terms and conditions of Waste Discharge Requirements Order No. 75-168. This discharge was supposed to be contained to land; however, increases in tailwater flow and storm water runoff from agricultural land upgradient of LNAS caused treated wastewater to discharge to the Kings River. Accordingly, the Regional Board adopted Order No. 82-034, which included a time schedule for the Discharger to implement corrective measures to eliminate the discharge of treated wastewater to the Kings River. To comply, the Discharger indicated that it would construct

facilities to convey wastes from the Operations Area to the Administrative Area WWTF. The Discharger continued to use its industrial WWTF for treating waste generated from washing aircrafts, but abandoned its Operations Area domestic WWTF and started piping raw domestic wastewater and industrial effluent from the Operations Area to the Administrative Area in 1984. Since 1984 the Discharger's treatment of domestic WWTF has gone unchanged, while it has made several modifications to its industrial WWTF to improve the quality of industrial effluent.

Wastewater Process Flow

Industrial wastewater is screened and then pumped to an oil and water separator from which it is pumped to a holding tank. From the holding tank, wastewater is pumped to one of two parallel treatment trains that consist of a second oil and water separator, alum mixing tank, polymer mixing tank, dissolved air flotation unit, lime mixing tank, a second polymer mixing tank, and a sedimentation basin. LNAS adds alum and polymers to enhance flocculation, and adds lime to remove metals. After the sedimentation basin, LNAS adjusts the pH of wastewater with sulfuric acid before discharging it to the Operations Area domestic sewage trunk line.

LNAS routes domestic wastewater through two “muffin monsters” then discharges it to two stabilization ponds (Ponds A and B), which operate in series and are designed to be facultative (i.e., wastewater is stratified into three layers: an anaerobic bottom layer, an aerobic surface layer, and an intermediate layer). In the middle and bottom of each stabilization pond there is a 400-foot square concrete structure into which influent is discharged. The concrete structures serve to accumulate sludge. From Pond B, LNAS has the option to discharge wastewater via sluice gates to one of three evaporation ponds. The ponds have the following dimensions:

<u>Pond Designation</u>	<u>Surface Area (acres)</u>	<u>Depth (feet)</u>	<u>Volume (acre-feet)</u>
Pond A	28	8	224
Pond B	18	8	144
Pond 1	116	10	1,160
Pond 2	100	10	1,000
Pond 3	81	10	810

Currently, LNAS discharges Pond B effluent to Pond 1. When Pond 1 nears its hydraulic capacity, LNAS discharges excess water from Pond 1 to Pond 2. The Discharger also recycles effluent via a recirculation pump from Pond 1 to Ponds A and B, as necessary, to control odors and provide dilution of wastewater entering the stabilization ponds. The Discharger does not currently discharge effluent directly to Pond 3. To ensure that effluent does not percolate to groundwater and adversely affect farming activities (i.e., create drainage problems), LNAS installed a leachate collection system around its ponds. The leachate collection system also serves to dewater nearby farmland. LNAS discharges collected leachate exclusively to Pond 3 for final disposal via evaporation.

Sludge Process Flow

From its industrial WWTF, the Discharger discharges sludge from the holding tank, first-phase oil and water separator, and sedimentation basin to a sludge holding tank, then to a 1,200-gallon metal diatomaceous earth tank before dewatering it with a pressure filter press. All leachate and decant are recycled back to the headworks of the industrial WWTF. The spent diatomaceous earth and sludge are landfilled at Chemical Waste Management's Kettleman Hills Facility, which the Regional Board regulates by Waste Discharge Requirements Order No. 98-058.

According to the Discharger's 1995 *Sewage Sludge Management Plan*, a survey crew made soundings of the top of the sludge layer in domestic WWTF ponds by using a prism attached to a rigid PVC pole. The resulting depths were compared with the as-built plans for the ponds to determine that about 0.5 feet of sludge had accumulated in each of the five ponds. This equates to a sludge build-up rate of about 0.3 inch/year. The Discharger's 1995 *Sewage Sludge Management Plan* also indicated that it would take sixty years for sludge to accumulate to levels that would affect the ponds' treatment process, as the WWTF's facultative ponds are 8 feet deep and capable of providing adequate treatment at a minimum depth of 4.5 feet.

Compliance Issues

Order No. 86-173 specifies that dissolved oxygen levels in domestic WWTF ponds shall not be less than 1 mg/L and that effluent from the industrial discharge shall (1) have a pH between 6.5 and 8.5 and (2) not exceed an EC of 1,000 μ mhos/cm, oil and grease of 15 mg/L, TSS of 60 mg/L, total cyanide of 0.43 mg/L, copper of 3.38 mg/L, nickel of 0.1 mg/L, total chromium of 0.5 mg/L, zinc of 2.61 mg/L, lead of 0.5 mg/L, and cadmium of 0.1 mg/L.

The Discharger has chronically violated its industrial effluent limitations for EC and oil and grease, and occasionally violated industrial effluent limitations for metals, total suspended solids, pH, and cyanide. The Discharger has also occasionally violated domestic WWTF pond limitations for dissolved oxygen. In addition, dichloromethane in industrial effluent has at times been discharged in concentrations (even when diluted by domestic wastewater) that threaten to exceed water quality objectives.

Dissolved Oxygen Violations

In 1989 and 1990, the Discharger attributed violations of pond dissolved oxygen specifications to inoperable pumps designed to recirculate water from Ponds 1 and 2 to Ponds A and B. The Discharger repaired the recirculation pumps in January 1990. When DO violations continued, the Discharger indicated that this is to be expected, as the stabilization ponds (Ponds A and B) operate similarly to an anaerobic digester, and higher DO levels may be detrimental to facultative treatment. To control odors from its ponds, the Discharger indicated that the most appropriate mechanism would be continued operation of recirculation pumps.

In 1990, the Discharger requested that the Regional Board consider changing the dissolved oxygen requirement from a daily minimum to a monthly average. The Department of Fish and Game (DFG) indicated that it was not appropriate for the Regional Board to grant this request. To support its position,

DFG indicated that in 1990 there was an outbreak of avian botulism (a microorganism that prefers anaerobic conditions) at the domestic WWTF ponds.

While Order No. 86-173 requires that the Discharger monitor dissolved oxygen levels between 0800 and 0900, the Discharger has not reported the time it collects samples in self-monitoring reports. The table below summarizes dissolved oxygen concentrations (in mg/L) from January 2000 through October 2001:

<u>Pond Designation</u>	<u>Average</u>	<u>Minimum</u>	<u>Maximum</u>
Pond 1	8.5	1.0	24.0
Pond 2	6.8	0.1	24.0
Pond 3	7.2	1.2	15.3
Pond A	3.8	0.0	21.9
Pond B	6.8	0.0	22.4

As shown above, the dissolved oxygen levels in all five ponds have varied significantly. The saturation value for oxygen in water between 10 and 20° Celsius ranges from 11.3 to 9.2 mg/L. If significant photosynthesis occurs (i.e., excessive algal growth), supersaturation of dissolved oxygen levels by as much as 250 percent can occur, according to *Limnology* by Alexander Horne and Charles Goldman. As the maximum results recorded by the Discharger indicate excessive algal growth, a diurnal variation in dissolved oxygen levels would be expected to occur in the five ponds. The appropriate mechanism for ensuring that nuisance conditions do not develop is to monitor dissolved oxygen in the early morning (the minimum level for a diurnal variation). As the Discharger has reported values nearly 2.5 times the saturation level, it suggests that these values were obtained by monitoring in the afternoon (the maximum level for a diurnal variation) instead of as directed by Order No. 86-173.

Oil and Grease Violations

Oil and grease (O&G) in industrial effluent has frequently exceeded 15 mg/L, the effluent limitation prescribed by Order No. 86-173. Beginning in 1995, the Discharger made several improvements and modifications to its treatment system to enhance O&G removal. These include the installation of new pumps to prevent oil from emulsifying in the treatment system, new screens, oil and water separator, and storage tank. Despite these efforts, the O&G in industrial WWTF still frequently exceeds 15 mg/L. However, as LNAS combines industrial effluent with domestic influent, which is treated in stabilization ponds and contained by a leachate collection system, the O&G concentration of combined industrial and domestic effluent is typically nondetect.

EC, Metals, TSS, and pH violations

The Discharger has occasionally violated its industrial effluent limitations for metals, TSS, and pH. Self-monitoring data from 2000 and 2001 do not show violations for TSS and only show slight violations for pH. Self-monitoring data for October 2001 indicated that the Discharger violated industrial effluent limitations for nickel and cadmium; however, recent self-monitoring data (2000 and 2001) do not indicate any other violations for these constituents.

Cyanide

The discharge exceeded the industrial effluent limits set forth in Order No. 86-173 for cyanide in June and July 1990; however, self-monitoring data from 2000 and 2001 only indicates one time when cyanide was detected in the discharge and the concentration was within the industrial effluent limit prescribed by Order No. 86-173.

Dichloromethane

In 1988 and 1989, industrial effluent contained high concentrations of dichloromethane (up to 71,000 µg/L). In 1996, industrial effluent contained levels of dichloromethane up to 920 µg/L. In 1997, the Discharger indicated that it had sealed all floor drains near a 1,450-gallon paint-stripping tank in building 170, where the majority of dichloromethane is used onsite. The Discharger reports that it removed the 1,450-gallon tank shortly thereafter. Self-monitoring data from 2000 and 2001 indicate that dichloromethane is normally nondetect, but detection limits for this constituent in self-monitoring reports have ranged significantly. It was detected in January 2001 (460 µg/L), June 2001 (5.5 µg/L), and December 2001 (6.8 µg/L). In summation, it appears that the Discharger has yet to remove all sources of dichloromethane from its discharge and continues to discharge this constituent at levels that may not represent best practicable treatment and control.

The Discharger indicates that it no longer has any tanks onsite that contain dichloromethane, only bottles such as cleaning paste and aerosol sprays. The Discharger reportedly uses dichloromethane in a manner that only generates dry waste (e.g., rags), according to its Standard Operating Procedures (SOP). To ensure that all onsite personnel are using dichloromethane in accordance with SOP, the Discharger indicates it will conduct a review of all onsite use of this chemical. In addition, the Discharger explains that this will be a continuous process, as new operators will need to be trained to ensure SOP are followed.

Agricultural Drainage Issues

During a 24 October 2001 inspection, Regional Board staff collected samples of influent (prior to the “muffin monsters”) and effluent (where Pond A discharges to Pond 1) at the domestic WWTF for EC, which indicated influent and effluent levels of 6,700 and 14,000 µmhos/cm, respectively. The Discharger attributed the high concentrations of EC to construction dewatering activities that it began in earnest around June 2001. The Discharger also indicated that the EC level of 6,700 µmhos/cm is suspect, as it is almost twice as high as any readings it has collected. The Discharger provided EC monitoring data it conducted with a hand-held meter from September 2001 through January 2002 that indicates EC levels have ranged from about 1,000 to 3,600 µmhos/cm. But, the EC reading of 6,700 µmhos/cm was accompanied by a standard minerals laboratory analysis that provided further evidence of high salinity, which makes it very unlikely that the EC reading was the result of a lab error. In addition, during a 15 August 2000 inspection, Regional Board staff collected samples of influent and at Pond A for EC, which indicated levels of 5,500 and 9,000 µmhos/cm, respectively. As Regional Board staff collected these samples before the Discharger initiated dewatering activities, it appears that tile drain water might be infiltrating into the Discharger’s collection system.

Carlton Duty, an adjacent landowner farming 240 acres north of the domestic WWTF, began discharging tile drain water into Pond 3 in 1983. By letter dated 20 April 1986, the Regional Board informed the Discharger that it was illegal to accept tile drain water from Mr. Duty, as it was not regulated by waste discharge requirements. In 1987 the Discharger stopped receiving tile drainage from Mr. Duty. Samples collected from March 1985 through June 1989 from evaporation Pond 3 revealed concentrations of selenium ranging from 11 to 32 µg/L. In response to a request by DFG, the Discharger collected additional samples for selenium from all its ponds in 1991 that revealed the following:

<u>Pond Designation</u>	<u>Number of Samples Collected</u>	<u>Concentration Range (µg/L)</u>
Pond 1	6	< 2
Pond 2	4	<2 to 3
Pond 3	4	4 to 6
Pond A	4	<2 to 2
Pond B	4	<2

Due to the elevated levels of selenium (i.e., above 2.7 µg/L) found in Pond 3, DFG recommended that the Discharger sample invertebrates for selenium. While the Discharger never sampled invertebrates for selenium, it did continue to monitor ponds for selenium. From 1998 through July 2000, the Discharger monitored Pond 3 for selenium, but concentrations were consistently nondetect. After collecting samples in July 2000, the Discharger changed laboratories. Since this time the Discharger has reported very high selenium concentrations in Ponds 2 and 3. Besides switching laboratories, recent construction activities that necessitate significant dewatering might also be responsible for high selenium concentrations, as the Discharger recently documented that shallow groundwater contains an average concentration of selenium of about 80 µg/L. The results from 2001 analyses are provided below:

<u>Pond Designation</u>	<u>Number of Samples Collected</u>	<u>Concentration Range (µg/L)</u>
Pond 1	4	9 to 10
Pond 2	4	210 to 220
Pond 3	8	200 to 300
Leachate Collection System	4	26 to 70

As shown above, selenium in the Discharger's ponds are at concentrations that can adversely impact waterfowl. The Discharger has indicated that laboratory results concerning selenium are inconclusive. To support its position, the Discharger collected split samples for selenium from Ponds A, B, 1, 2, 3, and its leachate collection system in January 2002. One laboratory reported concentrations of selenium ranging from 13 to 370 µg/L in the five ponds and leachate collection system, while the other reported that all values were nondetect (< 2 µg/L).

Hydrology and Soils

Regional groundwater flows southwesterly, according to information in *Lines of Equal Elevation of Water in Wells in Unconfined Aquifer*, published by the California Department of Water Resources in Spring 1998. Areal topography indicates a slope of about 1 foot per 1,000 feet toward the east. Surface

water drainage in the WWTF environs is by sheet flow to the Kings River, approximately two miles to the east close to where the North and Clarks Fork converge.

In the LNAS vicinity, three confining layers of clay significantly impede the migration of groundwater between aquifers: the A-Clay can be found at about 50 feet below ground surface (bgs), the C-Clay at about 225 feet bgs, and the E-Clay at about 650 feet bgs, according to *Final Basewide Groundwater Beneficial Use Technical Memorandum* (hereafter Report), dated 6 April 2001 and prepared by Tetra Tech EM Inc. for the U. S. Department of Navy. The Report indicates that the shallow-upper aquifer consists of water between 9 and 50 feet bgs, the intermediate-upper aquifer consists of water between 50 and 225 feet bgs, and the deep-upper aquifer consists of water between 225 and 650 feet bgs. Because extensive clay layers exist in the upper aquifer system, the permeability of area soils is considered low.

The Report evaluates water quality in the shallow-upper aquifer and concludes that it cannot potentially be used for either municipal or agricultural supply due to high levels of total dissolved solids. In addition, the Report provides Stiff diagrams from cluster wells that are screened at multiple intervals to show the geochemical difference in water quality below the shallow-upper aquifer to illustrate that the A-Clay limits the vertical transport of water. The Report indicates that the absence of agricultural wells in the shallow-upper and intermediate-upper aquifer and the high levels of salinity found in both these aquifers shows that they are not suitable for beneficial uses as a domestic, municipal or agricultural supply.

Annual average precipitation is about 8 inches. The average annual evapotranspiration rate in the LNAS vicinity is about 63 inches, according to the California Irrigation Management Information System Reference Evapotranspiration map. The pan evaporation rate is typically the evapotranspiration rate divided by 0.8, which, in this case, equates to about 79 inches/year. According to *Water in Environmental Planning*, by Thomas Dunne and Luna Leopold, the amount of evaporation from a pond is typically about 75 percent of the pan evaporation rate. Under this methodology, the evaporation rate from ponds would be about 59 inches/year.

Regional Board staff performed a water balance to quantify the amount of effluent that might percolate from the two stabilization ponds and three evaporation ponds, which total 340 acres. The average wastewater inflow to the domestic WWTF is about 1.9 mgd (2,130 acre-feet/year). Rainfall near LNAS totals about 8 inches/year, which equates to an annual inflow to the ponds of about 230 acre-feet. To determine the amount of evaporation from the ponds, an evaporation rate of 4.91 feet/year was used, which corresponds to 1,670 acre-feet/year of wastewater/effluent/rainfall evaporating from the ponds. As such, the remaining wastewater/effluent/rainfall (690 acre-feet/year) will percolate to groundwater (assuming the amount of water in ponds remains the same). If 690 acre-feet/year of wastewater/effluent/rainfall percolates to groundwater, this will result in a permeability of 2.0×10^{-6} cm/s.

It appears that a significant portion of the wastewater inflow to the domestic WWTF is tile drain water. Based on information provided in *Wastewater Engineering* by Metcalf & Eddy, Regional Board staff estimated that about 1.05 mgd (1,180 acre-feet/year) of wastewater originates from domestic sources. If the Discharger removes the tile drain component from its wastewater inflow, it could potentially recycle effluent on nearby cotton fields. To determine the water demand of cotton and hence the amount of

acres the Discharger could potentially irrigate with recycled water, Regional Board staff utilized the Blaney-Criddle Equation for calculating water requirements over a whole growing season:

$$E_t = [K * \sum (1.8 * T_{ai} + 32) * d_i] / 2.54 \text{ (from January through December)} \quad (1)$$

where: E_t = evapotranspiration (inches)
 K = crop coefficient for the whole growing season
 T_{ai} = average air temperature (°Celsius)
 d_i = the monthly fraction of annual hours of daylight

Regional Board staff followed the methodology outlined in *Water in Environmental Planning* by Thomas Dunne and Luna Leopold to determine cotton evapotranspiration values for each month. The result is an annual water demand for cotton of 45.11 inches (3.8 feet).

<u>Month</u>	<u>K (cotton)¹</u>	<u>(d_i)¹</u>	<u>(T_{ai})²</u>	<u>(E_t) (in)</u>
January	1.8	0.07189	7.78	2.34
February	1.8	0.06926	10.56	2.50
March	1.8	0.08363	12.78	3.26
April	1.8	0.08737	15.56	3.72
May	1.8	0.09574	20.00	4.61
June	1.8	0.095925	23.89	5.10
July	1.8	0.09774	26.67	5.54
August	1.8	0.09237	25.56	5.11
September	1.8	0.083	22.78	4.29
October	1.8	0.079445	18.33	3.66
November	1.8	0.071075	11.67	2.67
December	1.8	0.072555	7.22	2.31

¹ Thomas Dunne and Luna Leopold. *Water in Environmental Planning*. 1978. Total: 45.11

² Average Temperature from the National Weather Service website www.weather.com

If the annual water demand of cotton is 3.8 feet, the Discharger could irrigate about 310 acres per year with treated wastewater.

Land Use

The LNAS encompasses about 20 square miles. The City of Lemoore is about six miles east of the WWTF. A significant portion of the land at or near LNAS is used to grow crops by lease agreements with area farmers. Crops are predominantly irrigated with surface water. According to Land Use Maps prepared by the California Department of Water Resources in 1996, the predominant crop is cotton followed by corn, alfalfa, safflower, and pistachios. The LNAS is within the Westlands Water District.

Final Basewide Groundwater Beneficial Use Technical Memorandum (hereafter Report), dated 6 April 2001, and prepared by Tetra Tech EM Inc. for the U. S. Department of Navy indicates that the

Discharger identified the following sites in its industrial restoration program: (1) Landfill; (2) Pesticide Rinse Area-Building 752, (3) Pesticide Rinse Area-Building 50, (4) Old Fire Training Area, (5/9) Fire-fighting school/sludge drying ponds, (6) Operations Area Open Ditch, (7) Housing Area Open Ditch, (8) Housing Area sludge drying ponds, (10) Pesticide Application Landing Strip, (11) Transformer Oil Spills-Building 4, (12) Transformer Oil Spills-Building 468, (13) Transformer Storage Area-Building 50, (14) Solvent Groundwater Contamination, (15) Franklin Avenue-Building 50, (16) Sludge Ponds North of Building 65, and (17) jet petroleum (JP-5) Pipeline Fuel Leak. The Report indicates that the Discharger determined that no action was required for sites 3, 6, 7, 8, 10, 11, 12, 13, 15, and 16.

Sites Requiring Action

The Discharger operated a 39-acre unlined landfill that accepted about 34,000 cubic yards of solid waste per year (some may have been hazardous) from 1961 to 1989. In 1997, the Discharger capped the landfill. Currently, the Discharger is performing postclosure monitoring under the terms and conditions of Waste Discharge Requirements Order No. 5-00-091. The Discharger originally designated sites 2 and 4 as no action sites; however, the California Department of Toxic Substances Control (DTSC) required the Discharger to implement institutional controls at both of these sites.

The Report indicates that the Discharger conducted groundwater monitoring in the shallow-upper aquifer near sites 5/9 and 14. The Discharger found contamination at site 5/9 for chlorinated volatile organic compounds (VOCs), total petroleum hydrocarbon-extractable (TPH-e), and polynuclear aromatic hydrocarbons over an area covering 324,000 square feet. At site 14, the Discharger found contamination for chlorinated VOCs, TPH-e, and total petroleum hydrocarbon-purgeable. The Report indicates that the Discharger will continue monitoring at these sites to further assess the extent of contamination. While the Discharger has predominantly monitored the shallow-upper aquifer, the Report indicates that it collected some samples from the intermediate-upper aquifer, which revealed trace concentrations at sites 5/9 and 14 for the aforementioned constituents.

The Report also indicates that the Discharger continues to find floating product resulting from a JP-5 pipe leak at site 17. The Discharger has determined that contamination of groundwater in the shallow-upper aquifer covers an area of about 250,000 square feet. To remove floating product, the Discharger reportedly installed a multiphase extraction system in March 2001.

Sludge Drying Ponds

In 1961, the Discharger began discharging sludge generated by the industrial WWTF to two unlined ponds adjacent to the western boundary of the fire-fighting school. In 1981, the Discharger classified the sludge in the west pond as hazardous waste, and therefore, removed the sludge and five feet of topsoil. The Discharger continued to operate the east pond until 1987. Cease and Desist Order No. 86-174 required the Discharger to close the east pond. LNAS implemented a Regional Board-approved closure plan in 1987, removed 3,263 cubic yards of sludge and contaminated soil, and installed a filter press. Accordingly, on 29 January 1988, the Regional Board adopted Order No. 88-003 to rescind Cease and Desist Order No. 86-174.

Groundwater Monitoring

To determine if an aquifer is suitable as a municipal or domestic water supply, the EPA and Regional Board provide criteria that must apply to a potential water source before that water source can be given consideration for classification as an unsuitable municipal or domestic source. If the concentration of TDS in groundwater exceeds 10,000 mg/L (EPA) or 3,000 mg/L (Regional Board), then the Regional Board may consider the groundwater as unsuitable to serve as a drinking water source. The Report indicates that between 1991 and 2000, LNAS collected 550 samples in the shallow-upper aquifer for TDS. Laboratory results indicate that more than 97 percent of the samples exceeded the state criterion of 3,000 mg/L, of which more than 70 percent exceeded the federal criterion of 10,000 mg/L. As the levels of TDS in the shallow-upper aquifer will also prohibit plant growth, the Report concludes that the shallow-upper aquifer cannot potentially be used for municipal, domestic, or agricultural supply.

By letter dated 30 August 2001, the Discharger provided average concentrations for metals from four background wells identified in Attachment D. The results provided below are from 22 analyses for each constituent:

<u>Constituent</u>	<u>Average Concentration (µg/L)</u>	<u>Primary MCL¹ (µg/L)</u>	<u>Secondary MCL² (µg/L)</u>
Aluminum	2,250	1,000	200
Antimony	570	6	NA
Arsenic	8.8	50	NA
Barium	31.6	1,000	NA
Beryllium	ND ³	4	NA
Cadmium	ND	5	NA
Calcium	320,000	NA ⁴	NA
Chromium	17.8	50	NA
Cobalt	ND	NA	NA
Copper	7.0	1,300	1,000
Iron	2,600	NA	300
Lead	10.5	15	NA
Magnesium	440,000	NA	NA
Manganese	59.7	NA	50
Mercury	0.12	2	NA
Molybdenum	180	NA	NA
Nickel	22.6	100	NA
Potassium	4,500	NA	NA
Selenium	81.1	50	NA
Silver	ND	NA	100
Sodium	2,900,000	NA	NA
Thallium	ND	2	NA
Vanadium	9.2	NA	NA
Zinc	30.5	NA	5,000

¹ Title 22, CCR, sections 64431 and 64670

² Title 22, CCR, section 64449

³ Nondetect

⁴ Not Applicable

Basin Plan, Beneficial Uses, and Regulatory Considerations

The antidegradation directives of section 13000 of the California Water Code require that waters of the State that are better in quality than established water quality objectives be maintained “consistent with the maximum benefit to the people of the State.” Waters can be of high quality for some constituents or beneficial uses and not others. Policies and procedures for complying with this directive are set forth in the Basin Plan (including by reference State Water Board Resolution No. 68-16, “Statement of Policy With Respect to Maintaining High Quality Waters in California,” or the “Antidegradation” Policy).

In allowing a discharge, the Regional Board must comply with CWC section 13263 in setting appropriate conditions. The Regional Board is required, relative to the groundwater that may be affected by the discharge, to implement the Basin Plan and consider the beneficial uses to be protected along with the water quality objectives essential for that purpose. The Regional Board need not authorize the full utilization of the waste assimilation capacity of the groundwater (CWC section 13263(b)) and must consider other waste discharges and factors that affect that capacity. The applicable beneficial uses (agricultural, industrial, and domestic supply in this instance), procedure for application of water quality objectives, and the process for and factors to consider in allocating waste assimilation capacity are set forth in the Basin Plan.

The WWTF is in the Hanford-Lemoore Hydrologic Unit of the Tulare Lake Basin. The *Water Quality Control Plan for the Tulare Lake Basin, Second Edition* (Basin Plan), designates beneficial uses, establishes water quality objectives, and contains implementation plans and policies for all waters of the Basin. The Basin Plan identifies existing and potential beneficial uses of area groundwater as municipal and domestic supply, agricultural supply, industrial service supply, and industrial process supply. The Basin Plan identifies existing and potential beneficial uses of the Kings River from Peoples Weir to Stinson Weir on the North Fork and to Empire Weir No. 2 on the South Fork as agricultural, water contact recreation, noncontact water recreation, warm freshwater habitat, wildlife habitat, and groundwater recharge.

For wastewater treatment facilities that discharge to land in a manner that waste may infiltrate below the ground surface and degrade groundwater, the Basin Plan requires treatment systems with a design flow greater than 1 mgd to provide removal of 80 percent or reduction to 40 mg/L, whichever is more restrictive, of both five-day biochemical oxygen demand and suspended solids. The Basin Plan also requires that the maximum electrical conductivity of a discharge to land that has the potential to degrade groundwater be limited to 500 µmhos/cm above source water.

State Water Resources Control Board Resolution No. 88-63 *Sources of Drinking Water* policy indicates that all groundwater in the State, including the Tulare Lake Basin, is considered to be suitable, or potentially suitable, for municipal or domestic water supply, and should be so designated by the Regional Board with certain exceptions. One of those exceptions is for water where the TDS exceeds 3,000 mg/L and it is not reasonably expected to supply a public water system.

The Basin Plan identifies existing and potential beneficial uses of underlying groundwater as municipal, domestic, and agricultural supply. As described previously, shallow groundwater in the LNAS vicinity is not used as a municipal or domestic supply due to high concentrations of TDS. It also exceeds primary MCLs for aluminum, antimony, and selenium, and secondary MCLs for aluminum, iron and manganese. Since the Discharger disposes of treated wastewater by discharging it to evaporation ponds overlying low permeable soils and underlying groundwater is of poor quality (i.e., TDS exceeds 3,000 mg/L), it does not appear that the discharge has a reasonable potential to degrade shallow groundwater for salinity. Accordingly, the antidegradation provisions of Resolution 68-16 that could pertain to salinity waste constituents do not apply to this discharge.

The Basin Plan indicates that the current method of maintaining drained land in production is to capture and dispose of salts in evaporation ponds; however, it also encourages recycling of municipal wastewater and does not consider disposal by evaporation and percolation or discharge to surface waters a permanent disposal solution when the potential exists for recycling. While the discharge or infiltration of tile drain water into the Discharger's treatment system may not reduce the organic removal capacity, it will render effluent unsuitable for recycling. This is inconsistent with the Regional Board's recycling policies.

Title 27

Title 27, CCR, section 20005 et seq. ("Title 27"), contains regulations to address certain discharges to land that would cause degradation in violation of Resolution No. 68-16 or any water quality objectives or other provisions of the Basin Plan. Title 27 precludes the discharge of such wastes to groundwater by requiring compliance standards contained therein. Accordingly, Title 27 establishes a waste classification system, specifies siting and construction standards for full containment of classified waste, requires extensive monitoring of groundwater and the unsaturated zone for any indication of failure of containment, and specifies closure and post-closure maintenance requirements. Generally, no degradation of groundwater quality by any waste constituent is acceptable.

Where the nature of the waste and the underlying groundwater (and any surface water may be affected) are such that a discharge of waste would not cause degradation in violation of Resolution No. 68-16 or any water quality objectives or other provisions of the Basin Plan, then the Regional Board may exempt the discharge from the provisions of Title 27.

Discharges of domestic sewage and effluent can be treated and controlled to a degree that will not result in unreasonable degradation of groundwater. For this reason, they have been conditionally exempted from Title 27, except for residual sludge and solid waste generated as part of the treatment process [section 20090(a) of Title 27]. The condition requires that the discharge not result in violation of any water quality objective in groundwater.

Treatment and storage facilities for sludge that are part of the WWTF are considered exempt from Title 27 under section 20090(a), under the condition that the facilities not result in a violation of any water quality objective. However, residual sludge (for the purposes of the proposed order, sludge that will not be subjected to further treatment by the WWTF) is not exempt from Title 27. Solid waste (e.g.,

grit and screenings) that results from treatment of domestic sewage and industrial waste also is not exempt from Title 27. This residual sludge and solid waste are subject to the provisions of Title 27.

Accordingly, the municipal discharge of effluent and the operation of treatment or storage facilities associated with a municipal wastewater treatment facility can be allowed without requiring compliance with Title 27, but only if resulting degradation of groundwater is in accordance with the Basin Plan.

In addition, Title 27, section 20090(b), exempts drainage water and sediments if (1) the applicable regional board issued waste discharge requirements or waived such issuance, (2) the discharge complies with the applicable water quality control plan, and (3) the discharge does not need to be managed according to Title 22, CCR, Division 4.5, Chapter 11, as a hazardous waste.

Therefore, the test to determine compliance with the Basin Plan is based on the comparison of the quality of the wastewater and the quality of the underlying groundwater. A facility is not exempt from Title 27 if poor quality wastewater could impair excellent quality groundwater and cause nonconformance with the water quality objectives of the Basin Plan and impair beneficial uses.

As the Discharger disposes of wastewater in evaporation ponds, uses a leachate collection system, and underlying groundwater is of poor mineral quality (i.e., TDS is greater than 3,000 mg/L); it does not appear that the discharge has a reasonable potential to adversely affect groundwater. Hence, the Discharger, conditional upon full compliance with this Order, qualifies for an exemption from Title 27.

PROPOSED ORDER TERMS AND CONDITIONS

Discharge Prohibitions, Specifications and Provisions

The proposed Order would prohibit the discharge of wastes to surface waters and the discharge to land of hazardous or designated wastes and requires the Discharger to implement best practicable treatment and control measures for the subject wastewater.

The industrial discharge specifications in the proposed Order for settleable matter and TSS would be carried over from the current Order. The table below indicates the industrial discharge specifications prescribed by Order No. 86-173 for copper, nickel, total chromium, zinc, total cyanide, silver, lead, cadmium, trichloroethane, and dichloromethane, the primary and secondary (in parentheses) drinking water standards, the average value reported by the Discharger in self-monitoring reports, and the average concentration of the constituents in background groundwater monitoring wells:

<u>Constituent</u>	<u>Order No. 86-173</u>	<u>Primary and (Secondary) MCLs</u>	<u>Average Value¹</u>	<u>Background Wells</u>
Copper (mg/L)	3.38	(1.0) ²	< 0.01	0.007
Total Chromium (mg/L)	0.5	0.05 ³	< 0.02	0.0178
Nickel (mg/L)	0.1	0.1 ³	< 0.02	0.0226
Zinc (mg/L)	2.61	(5) ²	0.2	0.031
Cyanide (mg/L)	1.2	0.2 ³	< 0.01	NA ⁴
Silver (mg/L)	0.43	(0.1) ²	< 0.01	ND ⁵

<u>Constituent</u>	<u>Order No. 86-173</u>	<u>Primary and (Secondary) MCLs</u>	<u>Average Value¹</u>	<u>Background Wells</u>
Lead (mg/L)	0.5	0.015 ⁶	< 0.01	0.011
Cadmium (mg/L)	0.1	0.005 ³	< 0.002	ND ⁵
Trichloroethane ⁷ (mg/L)	3	0.005 ⁸	< 0.03 ⁹	NA ⁴
Dichloromethane (mg/L)	--	0.005 ⁸	<0.04 ¹⁰	<0.001

¹ Self-monitoring data from January 2000 through October 2001

² Title 22, CCR, section 64449, Table 64449-A

³ Title 22, CCR, section 64431, Table 64431-A

⁴ Not Analyzed

⁵ Nondetect

⁶ Title 22, CCR, section 64672.3

⁷ For both 1,1,1-Trichloroethane and 1,1,2-Trichloroethane

⁸ Title 22, CCR, section 64444, Table 64444-A

⁹ Average of detection limits, which ranged from 0.005 to 0.5 mg/L.

¹⁰ Average of detection limits (19) and reported values (3), detection limits ranged from 0.003 to 0.3 mg/L.

While shallow groundwater currently cannot be used for municipal or domestic supply due to high levels of TDS, background groundwater data does not indicate that any of the constituents described in the above table exceed drinking water standards. Any degradation of groundwater for waste constituents subject to treatment and control must be consistent with the Antidegradation Policy. State Water Resources Control Board Order No. WQ 91-10 indicates that high quality waters are determined based on specific properties and characteristics, and therefore, waters can be of high quality for some constituents, but not others. To ensure that the Discharger maintains the high quality of groundwater for the above constituents, and by extension, complies with Resolution 68-16 and 40 CFR Part 131.12(a)(2), the proposed Order would include industrial effluent limitations that will result in domestic WWTF effluent meeting the most stringent drinking water standard in Title 22. To determine the maximum permissible industrial effluent limitations, complete mixing of wastewater attributable to domestic sources (1.05 mgd) with the industrial WWTF design flow of 0.05 mgd was used. To ensure that the Discharger implements BPTC at its industrial WWTF, the proposed Order would contain industrial effluent limitations that equate to those prescribed by Order No. 86-173 or 21 times (1.05/0.05) the most stringent drinking water standard in Title 22, whichever is less.

The Oil and Grease violations section explains that the Discharger has implemented numerous measures that illustrate its resolve to comply with the oil and grease discharge specification contained in Order No. 86-173. To meet a daily maximum oil and grease limitation of 15 mg/L, the Discharger indicates it would incur significant expenses, as it would need to install carbon filters. Since domestic wastewater significantly dilutes industrial effluent (current dilution is over 100:1), the discharge is contained to land, and the Discharger appears to have implemented BPTC for limiting oil and grease in its discharge, the proposed Order would increase the industrial effluent limitation for oil and grease from a daily maximum of 15 mg/L to a 12-month moving average of 100 mg/L.

As domestic wastewater will effectively neutralize the pH of industrial effluent and the Discharger indicates that the piping containing only industrial effluent is compatible with high pH, the proposed Order would not carry over the industrial effluent pH limitation prescribed in Order No. 86-173.

While the Discharger chronically violated the effluent EC limitation of Order No. 86-173, the proposed Order would remove EC limitations for the discharge since (1) it is contained in evaporation ponds and (2) underlying groundwater is of poor mineral quality (i.e., TDS > 3,000 mg/L).

To ensure that nuisance conditions and conditions that are favorable to an outbreak of avian botulism do not develop at domestic WWTF ponds, the proposed Order would prescribe specifications for pond dissolved oxygen, water depth, freeboard, and vegetation management.

While the Discharger indicates that the high selenium concentrations found in its domestic WWTF ponds may be due to laboratory error, recent monitoring data suggests that waterfowl could be adversely affected. As the concentrations reported are at times orders of magnitude higher than the minimum levels that could impact waterfowl, the Discharger must implement wildlife monitoring until such time it can demonstrate that domestic WWTF ponds do not have the potential to impact wildlife. The proposed Order would require the Discharger to perform bird counts semimonthly throughout the year. To mitigate the impacts to birds, the proposed Order would require the Discharger to conduct a hazing program during the months of February through July that effectively keeps birds moving at any pond where birds are present and where the geometric mean selenium concentration in water is equal to or greater than 2.0 µg/L. The proposed Order would require the Discharger to submit technical reports regarding the concentrations of dichloromethane in industrial effluent, the effect of tile drain water on its treatment system (including an evaluation of the integrity of the sewer collection system), and the feasibility of recycling effluent on nearby farmland.

It may not be appropriate for the Discharger to land apply sludge that has accumulated in its ponds, as tile drain water may have caused the sludge to become unsuitable for use as fertilizer (i.e., increased levels of boron, sodium, chloride, and TDS to levels that are toxic to plants). The conditions for sludge, solid waste, and biosolids management contained in the proposed Order are intended to assure that degradation resulting from the Discharger's management of sludge would be in accordance with the Basin Plan. The proposed Order would require that storage, use and disposal of biosolids comply with the self-implementing federal regulations of 40 CFR 503, which are subject to enforcement by the U. S. Environmental Protection Agency not the Regional Board, and the Statewide General Order for the Discharge of Biosolids (Water Quality Order No. 2000-10-DWQ) (or any subsequent document which replaces Order No. 2000-10-DWQ).

Monitoring Requirements

Section 13267 of the CWC authorizes the Regional Board to require monitoring and technical reports as necessary to investigate the impact of a waste discharge on waters of the State. In recent years there has been increased emphasis on obtaining all necessary information, assuring the information is timely as well as representative and accurate, and thereby improving accountability of any discharger for meeting the conditions of discharge. Section 13268 of the CWC authorizes assessment civil administrative liability where appropriate.

To determine the efficiency of the Discharger's operation, the proposed Order would require the Discharger to monitor domestic WWTF influent weekly for settleable solids and pH; and monthly for BOD₅ and TSS. In order to adequately characterize domestic WWTF effluent, the Discharger would be required to monitor weekly for settleable solids and pH; monthly for EC, TSS, BOD₅, TDS, selenium and general minerals; semiannually for nitrogen constituents; and annually for antimony, molybdenum and arsenic. To monitor evaporation ponds for capacity constraints and potential nuisance conditions, the Discharger would be required to monitor freeboard available, water depth, and dissolved oxygen content weekly. To ensure that evaporation ponds are not having an adverse affect on wildlife, the proposed Order would require the Discharger to monitor selenium levels in each pond monthly. To ensure that the industrial WWTF operates as designed, the proposed Order would require the Discharger to monitor industrial effluent weekly for pH, EC, settleable solids, and TSS; twice/monthly for oil and grease and dichloromethane; monthly for copper, nickel, total chromium, zinc, silver, lead, cadmium, total cyanide, and trichloroethane; and annually for priority pollutants.

The proposed Order would require the Discharger to collect a composite sample of sludge from each domestic WWTF pond within 90 days of the adoption of the proposed Order, in accordance with EPA's *POTW SLUDGE SAMPLING AND ANALYSIS GUIDANCE DOCUMENT, AUGUST 1989*, and test for arsenic, cadmium, molybdenum, copper, lead, mercury, nickel, selenium, zinc, sulfate, boron, potassium, sodium, chloride, and EC. The proposed Order would also require the Discharger to submit an annual summary of sludge discharge operations from both its industrial and domestic WWTFs.

The proposed Order would also require that the Discharger inspect each WWTF pond weekly for dead birds or daily if (1) the water depth at any pond is less than two feet or (2) DFG determines that an outbreak of fowl cholera or botulism is occurring in the area. If, despite hazing, birds nest at the ponds, the proposed Order would require the Discharger to (1) perform breeding bird surveys semimonthly from March through July that include counts of nests, eggs, and young; (2) flag nests; and (3) collect random recurvirostrid egg samples from five separate nests and analyze for total recoverable selenium and moisture content.

The proposed action to revise waste discharge requirements for this existing facility is exempt from the provisions of the California Environmental Quality Act (CEQA) in accordance with Title 14, CCR, section 15302.

Reopener

The conditions of discharge in the proposed Order were developed based on currently available technical information and applicable water quality laws, regulations, policies, and plans, and are intended to assure conformance with them. It may be appropriate to reopen the Order if applicable laws and regulations change, but the mere possibility that such laws and regulations may change is not sufficient basis for reopening the Order. The CWC requires that waste discharge requirements implement all applicable requirements.